
LogicTree-RAG: Logic Tree-guided Retrieval-Augmented Generation for Long-form Patent Drafting

Jiaqi Zhu

National University of Singapore
jiaqi77@nus.edu.sg

Naili Xing

National University of Singapore
xingnl@comp.nus.edu.sg

Hexiang Pan

National University of Singapore
panh@u.nus.edu

Haotian Gao

National University of Singapore
gaohaotian@comp.nus.edu.sg

Jianwei Yin

Zhejiang University
zjuyjw@zju.edu.cn

Xiaokui Xiao

National University of Singapore
xkxiao@nus.edu.sg

Beng Chin Ooi

Zhejiang University
ooibc@zju.edu.cn

Abstract

Long-form technical text generation underpins knowledge-intensive workflows, yet remains challenging for large language models (LLMs) due to the need for globally consistent logical structuring and faithful technical reasoning beyond local coherence. Patent drafting is a canonical instance of this challenge, demanding holistic generation of a legally compliant and technically exhaustive document through sustained multi-expert collaboration. Existing approaches often focus on partial section generation or rely on manually crafted outlines, limiting scalable automation in realistic settings. In this work, we propose LogicTree-RAG, a logic tree-guided retrieval-augmented generation framework that induces a hierarchical logic tree as a global organizational backbone to organize and ground technical disclosures, without relying on expert-defined drafting priors. Each node in the logic tree represents a technical element and is constructed through evidence-guided recursive generation. A hybrid traversal mechanism then maps the logic tree into patent sections, enabling controllable and section-balanced generation. Extensive experiments show that LogicTree-RAG consistently improves content quality and language conformity over strong LLM-based baselines and achieves longer structured generation with high token efficiency, demonstrating the effectiveness of logic-centric generation for complex technical document drafting.

1 Introduction

Patents are a cornerstone of intellectual property protection, safeguarding technological innovations and enabling their translation into practical applications [1, 2]. Drafting a patent, however, is a highly specialized task that requires both a comprehensive understanding of the underlying technical contribution and a detailed knowledge of the formal and legal conventions of patent documents [3, 4]. In practice, this process often involves close collaboration between *inventors*, who possess the technical expertise but typically lack familiarity with patent law, and *patent attorneys*, who are well-versed in legal conventions but may not fully understand the technical intricacies of the invention. Consequently, such multi-expert collaboration often necessitates repeated discussion and iterative refinement, making the drafting process labor-intensive and time-consuming.

Recent advances in large language models (LLMs), which have demonstrated remarkable performance across a wide range of natural language processing (NLP) tasks, open up new possibilities for automating and accelerating patent drafting [5, 6, 7]. Prior work in this area has primarily focused on generating specific sections of patent applications [8, 9, 10, 11, 12, 13, 14], such as title, abstract, or claims, rather than producing a complete document. While useful for targeted drafting assistance, such partial generation fails to address the core challenge of producing a logically structured patent application in its entirety, where interdependencies across sections are essential.

While a few recent studies have explored full-patent generation, they typically rely on substantial manual pre-design and drafting efforts. For instance, AutoPatent [15] presupposes user-provided draft information, while Pap2Pat [16] assumes a detailed outline of each patent section written in advance. In their task setups, both methods approximate such user-provided inputs by extracting supervision from reference patents, which distances their designs from real-world patent drafting conditions and ultimately limits automation and scalability.

To address the above-mentioned problems, we formulate patent drafting as a long-text generation task, focusing on producing a complete and logically structured patent application from a pre-publication research paper serving as a proxy for the invention report (IR) [16]. This formulation captures the practical scenario in many research environments, where research papers frequently serve as the initial disclosure for patent drafting. Despite its practical relevance, advancing toward fully automated patent drafting entails several non-trivial challenges. First, patent applications are inherently long-form technical documents, requiring the generation of substantially long and complete texts that remain challenging for LLMs to sustain [17, 18]. Second, beyond producing long outputs, effective patent drafting demands globally consistent logical organization and balanced elaboration, a constraint that becomes increasingly difficult to maintain as generation length grows. Third, robust automation requires minimizing dependence on manually crafted priors, as such reliance limits scalability and generalization across drafting scenarios.

To tackle these challenges, we propose LogicTree-RAG, a logic tree-guided retrieval-augmented generation framework for long-form patent drafting. Unlike prior methods that depend on flat-generation LLMs with manually crafted outlines or templates, LogicTree-RAG introduces a structured generative paradigm in which *logic precedes content*: it automatically derives a hierarchical logic tree from the input research paper as a global organizational backbone and progressively transforms it into well-formed patent sections in a principled and controllable manner.

As the first step, LogicTree-RAG incorporates a semantic chunking mechanism that structures the input document into semantically coherent units, which are embedded into a vectorized document knowledge base. Upon this foundation, we introduce an evidence-guided recursive generation process for constructing a logic tree that captures the conceptual structure of the underlying research. This process leverages a novel node-aware semantic search strategy equipped with complementary expansion and refinement procedures, enabling section-level coherence and structural balance. Each node in the tree corresponds to a technical element, such as a principle, subtask, or subsystem, and is dynamically grounded by evidence retrieved from the knowledge base. Finally, a hybrid traversal strategy transforms the completed logic tree into patent sections, yielding a long-form, logically organized patent draft without requiring expert-defined priors. By serving as a logic-centric intermediate representation, logic tree bridges factual evidence with global generative control, enabling scalable long-form patent drafting.

We summarize our main contributions as follows:

- We propose LogicTree-RAG, a logic-centric generative framework for research-to-patent transformation that produces long-form, logically structured patent drafts without reliance on expert-defined drafting priors.
- We present an evidence-guided recursive generation approach that constructs a logic tree to model the conceptual dependencies and technical contributions with explicit factual grounding in the source document.
- We introduce a node-aware semantic search with a discriminative relevance scoring that jointly accounts for semantic similarity and structural distinctiveness, improving balanced elaboration and global structural coherence in patent drafting.
- Extensive experiments under a curated suite of evaluation metrics demonstrate that LogicTree-RAG improves technical completeness by 11.03% and content fidelity by 6.33% over strong LLM-based baselines, while enabling longer structured generation with high token efficiency.

2 Preliminaries

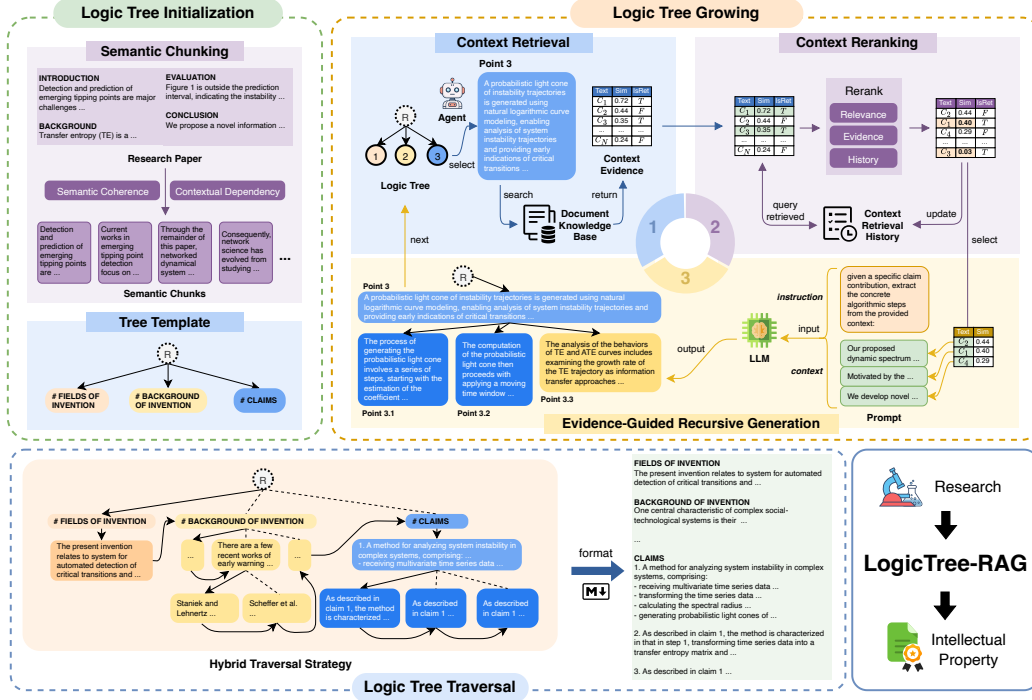


Figure 1: Architecture of LogicTree-RAG framework for long-form patent drafting.

Patents and Patent-Paper Pairs. A patent is a structured legal-technical document, typically consisting of *claims*, which define the legally enforceable scope of protection, and a *description*, which discloses the invention in sections such as Background, Summary, and Detailed Description [19, 16]. In many research domains, patents are filed alongside scientific publications, producing *Patent-Paper Pairs* [20, 21, 22, 23], where a pre-publication paper serves as the IR for the patent. Patent-paper pairs align scientific discourse with legal-technical documentation, offering a valuable yet underexplored resource for studying long-form technical text generation with LLMs.

Problem Formulation. Consider a patent-paper pair (X, P) , where $X = \{x_1, x_2, \dots, x_m\}$ is a pre-publication research paper represented as a collection of text spans, and $P = \{p_1, p_2, \dots, p_n\}$ is the corresponding patent document. In this work, we formalize the task as generating patent P from paper X mediated by a logic tree $\mathcal{T}$ with evidence grounding, defined as follows.

Definition 2.1 (Logic Tree). A logic tree is a directed rooted tree $\mathcal{T} = (V, E, r)$, where $V = \{v_1, \dots, v_N\}$ is the set of nodes, $E \subseteq V \times V$ the set of directed edges specifying parent-child relations, and $r \in V$ the root node representing the overall description of the document. Each node $v \in V$ is associated with two attributes: (i) a generated text fragment $c(v)$ and (ii) a set of supporting evidence spans $\mathcal{E}(v) \subseteq X$.

Intuitively, the root node summarizes the invention, internal nodes correspond to higher-level sections, and leaf nodes expand into fine-grained technical details. To ensure factual grounding, we impose the attribution constraint

$$\forall v \in V, \quad \mathcal{E}(v) \neq \emptyset, \quad (1)$$

which requires each generated unit to be anchored to at least one evidence span in the source paper.

3 Methodology

3.1 LogicTree-RAG Overview

As illustrated in Figure 1, LogicTree-RAG functions as a top-down pipeline that first parses the source document into structured textual units and constructs a logic tree that captures the conceptual

dependencies. Each node in the tree represents a technical element, such as a principle, step, or subsystem, and is dynamically associated with retrieved evidence from a vectorized document knowledge base. LogicTree-RAG then iteratively expands and refines the tree via evidence-guided recursive generation, supported by node-aware semantic search to ensure precise evidence alignment and structural consistency. Finally, the framework traverses the completed logic tree and hierarchically aggregates node contents into patent sections, yielding a long-form, section-balanced patent draft that integrates technical fidelity with structured generative controllability.

3.2 Hierarchical Logic Tree Initialization

To initialize retrievable knowledge units, we utilize layout-aware document analysis [24] to identify hierarchical boundaries across sections and paragraphs, reconstructing the document’s logical structure. Subsequently, we perform semantic chunking to produce contextually coherent segments that from the basis of the vectorized document knowledge base $\mathcal{V}_{kb}$. Unlike fixed chunking methods [25, 26, 27, 16], our approach dynamically determines chunk boundaries according to the semantic coherence and contextual dependency between adjacent paragraphs. Formally, given an ordered sequence of paragraphs $X = \{x_1, x_2, \dots, x_m\}$, we compute the pairwise semantic similarity:

$$s_{i,i+1} = \text{sim}(\mathcal{E}_p(x_i), \mathcal{E}_p(x_{i+1})), i = 1, \dots, m - 1, \quad (2)$$

where $\mathcal{E}_p(\cdot)$ denotes the paragraph-level embedding function, and $\text{sim}(\cdot, \cdot)$ is instantiated with cosine similarity. Paragraphs are iteratively merged when $s_{i,i+1} \geq \tau_s$ and the accumulated token length does not exceed the upper bound t_{max} . This procedure yields a set of semantically coherent chunks $\tilde{X} = \{\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_L\}$ that align with the recovered hierarchy. Details are provided in Appendix C.

Following semantic chunking, we initialize a logic tree $\mathcal{T} = (V, E, r)$ to organize and drive the generation process. To construct an initial structural scaffold, we prompt the backbone LLM with a high-level instruction σ_I and the chunk set $\tilde{X}$, generating a draft technical description $\mathcal{D}_d = \mathcal{M}_{LLM}(\sigma_I, \tilde{X})$. The root node r is instantiated with $c(r) = \mathcal{D}_d$ and $\mathcal{E}(r) = \tilde{X}$, representing the target patent document. Instead of utilizing syntactic rules or heuristics, we leverage $\mathcal{M}_{LLM}$ to induce the first layer of the logic tree by producing an evidence-grounded decomposition of key technical components from $\mathcal{D}_d$. This yields an ordered tree structure in which the root summarizes the invention and its first-level children nodes define the core semantic dimensions to be recursively expanded in subsequent stages.

3.3 Evidence-Guided Recursive Generation

Given the initialized logic tree $\mathcal{T}$, we perform recursive generation through a breadth-first expansion strategy. Specifically, we maintain a FIFO search queue $\mathcal{Q}$ initialized with the leaf nodes of $\mathcal{T}$, ensuring the expansion proceeds in a top-down manner. For each iteration, we dequeue the front node v_t , and retrieve its parent node v_f from $\mathcal{T}$. The goal of recursive generation is to determine whether v_t should be expanded, refined, or skipped based on the semantic evidence retrieved from the document knowledge base $\mathcal{V}_{kb}$, and to perform the corresponding updates to the logic tree.

To obtain candidate evidence, we perform retrieval over $\mathcal{V}_{kb}$ and collect a set of context spans $\mathcal{D}_{ctx}$, referred to as context evidence, which are semantically aligned with the node’s content $c(v_t)$. A global novelty test is then designed to verify whether $\mathcal{D}_{ctx}$ is already covered by other nodes in the logic tree, excluding the child nodes of v_f . If no novel evidence is identified, v_t is skipped; otherwise, the logic tree is updated via two complementary procedures:

- Expansion procedure $E(\cdot)$: Conditioned on $\mathcal{D}_{ctx}$ and an expansion instruction σ_E , the model generates a supplementary paragraph, which is then segmented into coherent semantic units. Each unit is instantiated as a child node of v_t , forming the set $V_t^{child} = \{v_{tc1}, \dots, v_{tck}\}$, where $c(v_{tci})$ is initialized as the corresponding segmented content and $\mathcal{E}(v_{tci}) = \mathcal{D}_{ctx}$. All nodes in V_t^{child} are then enqueued into $\mathcal{Q}$ for further recursion.
- Refinement procedure $R(\cdot)$: The model examines $c(v_t)$ to identify under-specified terms using instruction σ_{R1} . For domain-specific concepts appearing in $\tilde{X}$, refinement is performed via retrieval-augmented generation with supporting evidence from $\mathcal{V}_{kb}$ and instruction σ_{R2} ; otherwise, the model completes the term with instruction σ_{R3} based on its internal knowledge.

Overall, the logic tree $\mathcal{T}$ is updated after each expansion or refinement step. The expansion procedure generates a set of child nodes based on the context evidence, while the refinement procedure resolves

under-specified terms and updates the content of the current node, expressed as:

$$E(\sigma_E, \mathcal{D}_{ctx}) = V_t^{child} \quad (3)$$

$$c(v_t) \leftarrow R(\sigma_R, c(v_t), \mathcal{V}_{kb}) \quad (4)$$

where $\sigma_R = \{\sigma_{R1}, \sigma_{R2}, \sigma_{R3}\}$. The recursive process terminates once either the queue becomes empty or the accumulated content length reaches a predefined budget, ensuring controllable expansion and preventing over-generation.

3.4 Node-Aware Semantic Search

During the evidence-guided recursive generation, the quality of the context evidence $\mathcal{D}_{ctx}$ critically influences the structural evolution of the logic tree. Naive retrieval may introduce redundancy or topic drift, particularly when the retrieved spans overlap with already-processed sibling nodes, leading to incoherent or uneven expansion. To ensure precision and structural consistency, we impose two desiderata for the context set: (i) strong semantic association with the current node, and (ii) low redundancy with its sibling nodes under the same parent. To meet these requirements, we introduce a node-aware semantic search mechanism. As illustrated in Figure 2, we first retrieve an initial set of candidate context spans $\tilde{\mathcal{D}}_{ctx}$ from $\mathcal{V}_{kb}$ using the current node content $c(v_t)$ as query:

$$\tilde{\mathcal{D}}_{ctx} = \mathcal{R}(c(v_t), \mathcal{V}_{kb}) = \text{Top-K}_{a \in \mathcal{V}_{kb}} \text{sim}(a, c(v_t)), \quad (5)$$

To jointly account for semantic specificity and structural discrimination, we evaluate each candidate span $d \in \tilde{\mathcal{D}}_{ctx}$ through a discriminative relevance score $S_{dr}(\cdot)$, defined as:

$$S_{dr}(d) = \text{sim}(d, c(v_t)) - \max_{v_e \in V_f^{child} \setminus \{v_t\}} \text{sim}(d, c(v_e)), \quad (6)$$

where V_f^{child} denotes the set of child nodes of v_f and $\setminus$ indicates exclusion. A span d is retained only if $S_{dr}(d)$ exceeds a predefined threshold. The final context evidence is obtained as $\mathcal{D}_{ctx} = \{d \mid S_{dr}(d) \geq \tau_f\}$. The node-aware semantic search ensures that the retrieved evidence is both node-specific and structurally non-redundant, supporting focused and coherent recursive generation.

Remark. Although both the node-aware semantic search and the global novelty test operate on evidence relevance and redundancy, they address complementary levels of semantic discrimination. Node-aware semantic search enforces specificity at the *local sibling level* by filtering evidence overlapping with existing child nodes of v_t , while the global novelty test verifies coverage at the *global tree-wide level* by checking whether $\mathcal{D}_{ctx}$ has already appeared elsewhere in the logic tree outside the v_f branch. Together, these mechanisms provide coordinated local discrimination and global redundancy control, enabling coherent and non-repetitive logic tree construction.

3.5 LogicTree-RAG Workflow for Patent Generation

As outlined in Algorithm 1, LogicTree-RAG operates in two stages: (i) *logic tree construction*, which provides a structured and factually grounded representation of the source document’s conceptual space, and (ii) *patent drafting*, which systematically traverses the constructed logic tree to generate the final patent draft. The first stage yields a logic tree $\mathcal{T}$ that associates each node $v \in V$ with both generated content $c(v)$ and grounded evidence $\mathcal{E}(v) \subseteq X$, as formalized in Algorithm 2 in Appendix D, forming the structural foundation for patent drafting.

In the second stage, LogicTree-RAG generates the patent draft by traversing the logic tree $\mathcal{T}$ and aggregating node contents. The hierarchical structure of the logic tree naturally aligns with the organization of complex technical documents, such as patents, where high-level nodes capture

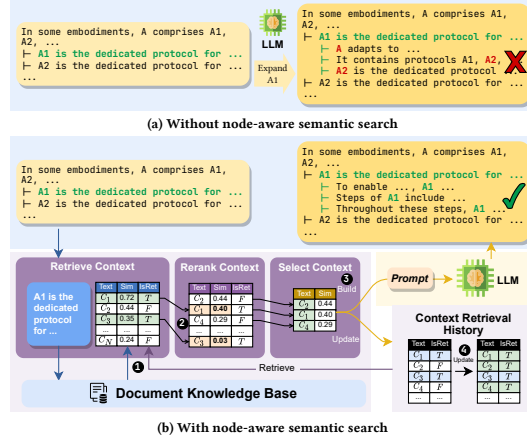


Figure 2: Generation without and with node-aware semantic search.

Table 1: Overall performance comparison of long-form patent drafting. The best results are highlighted in **bold**, and the second-best results are in **bold**.

Method	Output Tokens	Content-level				Language-level			Time (s) / Patent (Token)
		Coverage $\uparrow$	Factuality $\uparrow$		Semantic Similarity $\uparrow$	Style $\uparrow$	Repetition $\mathcal{R}_r \downarrow$	Coherence $\uparrow$	
			$\mathcal{F}_{Pat}$	$\mathcal{F}_{Src}$					
Heuristic Skyline									
Reference Patent (GT)	18.16k	89.05	88.74	88.80	100	100	13.91	100	-
Source Paper	8.03k	44.77	46.06	88.68	39.37	39.26	8.39	98.60	-
Single LLM-Call Generation									
Kimi K2	3.26k	38.21	42.76	58.95	21.28	39.47	12.04	95.92	64 (19.63)
Qwen3-80B	8.91k	39.09	37.50	44.88	34.08	35.06	6.05	95.71	81 (9.10)
Qwen3-Max	8.91k	39.44	37.68	45.22	34.09	35.06	8.41	97.14	172 (19.29)
DeepseekV3	1.87k	36.70	47.09	59.66	11.98	32.68	6.13	95.56	43 (22.89)
GPT-5 DT	2.95k	40.91	46.70	62.04	20.56	41.52	7.25	97.34	146 (49.49)
Gemini-2.5 Pro DT	2.69k	38.69	47.58	60.97	24.25	39.68	6.09	97.12	109 (40.52)
Agentic Framework Generation									
COPGEN	7.69k	39.21	57.63	62.39	32.56	59.54	15.47	97.11	265 (34.46)
COPGEN [†]	8.96k	32.85	52.44	58.16	28.96	43.38	18.94	97.21	216 (24.11)
$\hookrightarrow$ w/ SFT	25.85k	40.77	50.12	59.69	38.72	62.21	21.58	97.50	221 (8.55)
LongWriter	10.81k	37.21	36.65	45.21	21.12	34.26	6.67	95.62	197 (18.22)
LongWriter-Zero	12.16k	36.92	38.36	46.17	22.86	36.72	5.76	96.84	258 (21.11)
LogicTree-RAG	20.22k	43.79	58.92	66.34	38.91	65.68	4.14	97.45	311 (15.38)
$\hookrightarrow$ Contamination Control	18.96k	42.95	57.79	66.57	40.28	64.28	4.60	97.23	298 (15.72)

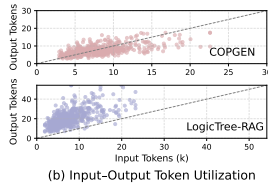
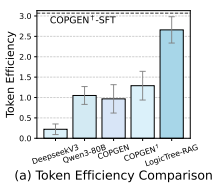


Figure 3: Token efficiency comparison results.

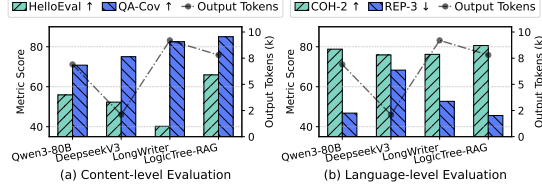


Figure 4: Generalization performance on LCFO.

under identical LLM decoding configurations to ensure a fair comparison. Detailed prompts and implementation settings are provided in Appendices E and F.

Evaluation Metrics. We evaluate patent generation from both content-level and language-level perspectives, complemented by a human expert study conducted by patent practitioners. *Content-level* metrics assess whether the generated patent conveys accurate and sufficient technical information grounded in the reference materials, including Coverage (NLI-based SCALE [38]), Factuality (patent-grounded $\mathcal{F}_{Pat}$ and source-grounded $\mathcal{F}_{Src}$), and Semantic Similarity (ROUGE-L [39]). *Language-level* metrics evaluate adherence to patent writing conventions, including Style (StyloMetrix [40]), Repetition ($\mathcal{R}_r$ [41]), and Coherence (DiscoScore [42]). Details are provided in Appendix F.1.

4.2 Main Results

The overall results are summarized in Table 1. LogicTree-RAG delivers consistently superior performance across key content and language metrics, generating substantially longer patent drafts that outperform strong long-context baselines. At the *content level*, LogicTree-RAG achieves the highest coverage, factuality, and semantic similarity among all baselines. These results underscore that logic-tree-guided generation improves technical completeness and evidence-grounded elaboration beyond single-call long-context prompting and agentic frameworks even with post-training. At the *language level*, LogicTree-RAG demonstrates strong performance in stylistic conformity and repetition rate. Its coherence score ranks second only to the SFT-tuned COPGEN variant, while requiring no task-specific fine-tuning. To control for potential training-data contamination, we evaluate LogicTree-RAG on a temporally filtered post-2024 subset beyond typical pretraining cutoffs, observing consistent gains without degradation, suggesting that the improvements are not driven by training-data overlap. Moreover, LogicTree-RAG maintains competitive per-token efficiency with only modest overhead, offering a favorable quality–efficiency trade-off for long-form patent drafting.

4.3 In-Depth Analysis of Patent Generation

Token Efficiency. We further analyze token efficiency, defined as the ratio of generated output tokens to total input tokens consumed across all LLM invocations, which serves as a proxy for both generation efficiency and scalability. As illustrated in Figure 3(a), LogicTree-RAG achieves the highest token efficiency among all baselines, closely approaching the SFT-enhanced COPGEN variant

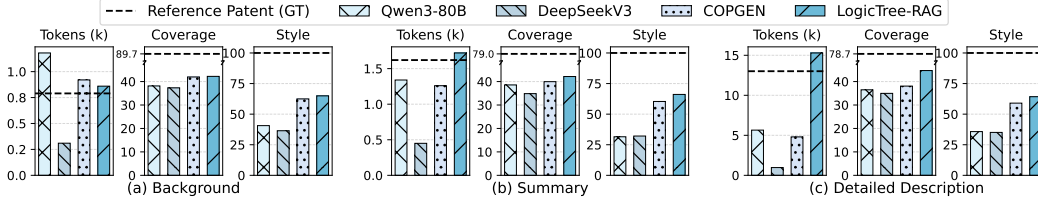


Figure 5: Section-level performance in terms of output length, content coverage, and linguistic style.

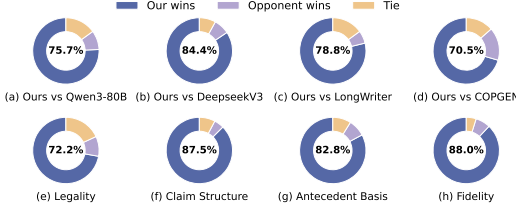


Figure 6: Win-rate results (%) under pairwise comparison.

Table 2: Component-wise ablation of LogicTree-RAG.

Variant	Output Tokens	Content-level		Language-level	
		Coverage	Factuality (F_{Pat})	Style	RR (R_+)
w/o Logic Tree	9.10k	39.85	56.92	60.47	16.86
Logic Tree w/o Retrieval	8.21k	40.56	51.16	59.05	9.38
w/o Semantic Chunking	19.42k	42.09	53.74	62.27	10.93
w/o Refinement	18.58k	39.92	56.85	61.87	4.09
w/o Evidence-Grounded σ_{R2}	19.60k	40.16	56.98	62.64	4.67
w/o Internal Knowledge σ_{R3}	18.95k	43.58	58.85	65.62	5.21
w/o Node-aware Reranking	20.85k	42.60	57.13	64.79	8.32
w/ DFS Traversal	21.13k	41.15	58.68	65.12	5.34
LogicTree-RAG	20.22k	43.79	58.92	65.68	4.14

despite requiring no task-specific fine-tuning. Figure 3(b) further shows that LogicTree-RAG shifts the input-output token distribution toward a more efficient regime and effectively converts contextual input into useful long-form patent content. Detailed cost breakdowns are provided in Appendix G.4.

Generalizability to Long-Text Generation. LogicTree-RAG is not tied to patent drafting, but targets a broader setting of source-grounded structured long-form generation, where the goal is to transform a long source document into a coherent and factually grounded target document. To validate this generalizability, we evaluate LogicTree-RAG on the LCFO benchmark [28] for document expansion. Following [28], we report output tokens, coherence (COH-2), repetition (REP-3), overall quality via an LLM-as-a-judge metric (HelloEval), and human-evaluated QA-based coverage (QA-Cov), which measures the fraction of benchmark questions answerable from the generated output. Without modifying the core pipeline, we apply LogicTree-RAG by adapting only the output schema. As shown in Figure 4, LogicTree-RAG consistently outperforms baselines in all content and language-level metrics, while maintaining competitive output length. These findings suggest that the benefits of logic-tree-guided generation extend beyond patent drafting to general long-form generation tasks.

Section-Level Analysis. We analyze generation quality across patent sections, i.e., Background, Summary, and Detailed Description, considering length, content coverage, and linguistic style, with results summarized in Figure 5. LogicTree-RAG produces section-wise output lengths closely aligned with ground-truth patents, indicating effective control over generation granularity and mitigating the length imbalance observed in single-call and chunk-based baselines. From a content perspective, LogicTree-RAG achieves the highest coverage across all patent sections, with up to a 17.4% improvement in the Detailed Description, where fine-grained technical completeness is critical and competing methods tend to under-generate. For language quality, LogicTree-RAG consistently attains the best style scores, demonstrating strong stylistic consistency across sections.

Human Expert Review. To assess practical applicability beyond automatic metrics, we conduct a blinded human expert evaluation with three patent practitioners using five criteria (i.e., Overall Quality, Legality, Claim Structure, Antecedent Basis, and Fidelity). Experts perform pairwise comparisons between our method and baselines. Figure 6(a-d) present pairwise comparisons against individual baselines, while (e-h) summarize performance across evaluation criteria. The results show that LogicTree-RAG consistently achieves strong human preference across all dimensions, indicating its practical utility in real-world scenarios. Full experimental details are provided in Appendix G.5.

4.4 Ablation and Sensitivity Analysis

Component-wise Ablation. We evaluate ablated variants of LogicTree-RAG with all other settings fixed, as shown in Table 2. Removing the logic tree or retrieval substantially shortens the outputs and degrades content quality, causing a 9.0% drop in coverage and a 13.2% degradation in factuality, respectively. This confirms the necessity of structural organization and evidence grounding. Removing semantic chunking or node-aware reranking weakens factuality and increases repetition, highlighting the importance of coherent and non-redundant evidence selection. Moreover, removing evidence-grounded refinement σ_{R2} leads to consistent degradation across all metrics, whereas removing

internal-knowledge refinement σ_{R3} yields negligible impact. Since σ_{R3} is triggered in only 8.6% of steps, it mainly serves as a fallback for under-specified cases, while the main gains come from retrieved evidence. Replacing hybrid traversal with pure DFS produces longer but less comprehensive outputs, indicating that the hybrid strategy better balances breadth and depth. Overall, the full LogicTree-RAG achieves the best performance across metrics, validating the joint contribution of structural modeling, evidence grounding, and traversal design.

Sensitivity to Key Hyperparameters. We study the effects of the semantic chunking threshold τ_s and the discriminative relevance threshold controlled by the ranked candidate k . As shown in Figure 7(a), both coverage and factuality improve as τ_s increases from low to moderate values, and peak around 0.7–0.8, suggesting that overly coarse chunks introduce noisy evidence, while overly fine-grained chunks reduce contextual sufficiency. Figure 7(b) shows that increasing k initially improves coverage by allowing more supporting evidence, but factuality peaks at a moderate value and slightly declines when weaker or redundant evidence is included. This highlights the importance of balancing evidence sufficiency and discriminative filtering, and supports the default setting of $k = 5$ as an effective compromise. Further analysis of tree growth budget parameters is provided in Appendix G.6.

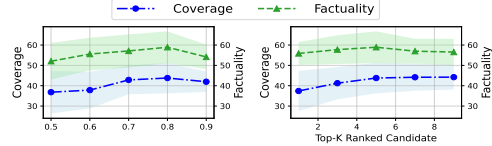


Figure 7: Hyperparameter Sensitivity Analysis.

We provide additional analyses on backbone choices, agentic pipeline variants, and evidence support in Appendices G.1, G.2, and G.3, further supporting the generality and reliability of LogicTree-RAG.

4.5 Broader Implications and Limitations

This work advances controllable and evidence-grounded generative AI for long-form technical document drafting. By framing patent drafting as a representative high-stakes structured generation task, LogicTree-RAG can assist practitioners in organizing complex technical disclosures into coherent and structured drafts, potentially reducing manual effort in early-stage drafting. However, the framework is intended to assist rather than replace domain experts, and its outputs should be treated as preliminary drafts and reviewed by qualified professionals before any formal use. Beyond patents, the logic-centric paradigm extends naturally to broader long-text generation tasks that require global organization and factual grounding. While LCFO results support such generalization, further validation on diverse long-form generation settings remains an important direction for future work.

5 Related Work

Patent-related Benchmarks. Recent progress in LLMs has spurred benchmarking of patent-centric capabilities [43]. For example, IPEval [44] evaluates patent law QA, PatentEval [12] targets claims-to-abstract and next-claim generation, and HUPD-DCG [45] measures claim generation from patent descriptions. Complementary to task-specific benchmarks, Pap2Pat [16] extends evaluation to long-text patent drafting, reflecting growing interest in document-level generation. These benchmarks illustrate the evolution of patent-related evaluation from localized capabilities toward holistic long-form technical writing.

Patent-related Generation. Existing patent generation methods can be broadly grouped into three paradigms. (i) *Patent-to-patent* generates specific sections conditioned on other parts within the same patent. For instance, PGT [8] is pre-trained to support title-to-abstract and abstract-to-claim generation. PatentLMM [46] presents a large multimodal model to write descriptions of patent figures. [13] proposes a knowledge-tuning approach for initial concept discovery and content generation. (ii) *Draft-to-patent* expands patent drafts into patent-form text. PatentGPT [11] pre-trains a series of base LLMs and benchmarks them with patent generation from technical disclosure documents draft. AutoPatent [15] designs a multi-agent workflow to migrate a form of answers to technical questions to the corresponding patent draft. (iii) *Paper-to-patent* targets long-form patent drafting from research papers, exemplified by Pap2Pat [16] with outline-guided generation. However, prior methods either operate under partial-context settings or depend on externally provided (often manually crafted) drafting priors, leaving realistic long-form patent drafting with global logical organization less well addressed. See Appendix B for a comprehensive review.

6 Conclusions

Automated patent drafting exposes a fundamental bottleneck in long-form technical text generation, where sustaining globally consistent logical structure and faithful technical reasoning remains challenging. In this paper, we propose LogicTree-RAG, a logic tree-guided retrieval-augmented generation framework for long-form patent drafting that introduces a hierarchical logic tree as a global organizational backbone, where each node represents a technical element and is grounded by retrieved evidence via an evidence-guided recursive generation mechanism. To ensure precise and balanced tree construction, we develop node-aware semantic search with discriminative relevance scoring and design a hybrid traversal strategy that maps the logic tree into structurally coherent patent sections. Extensive experiments show that LogicTree-RAG consistently improves content quality and language conformity over strong long-context and multi-stage LLM-based baselines, enabling longer structured generation with high token efficiency.

References

- [1] Edwin Mansfield. Patents and innovation: an empirical study. *Management science*, 32(2):173–181, 1986.
- [2] Christopher M Kalanje. Role of intellectual property in innovation and new product development. *World Intellectual Property Organization*, 2006.
- [3] Avery N Goldstein. *Patent Laws for Scientists and Engineers*. CRC Press, 2005.
- [4] Charlotte Waelde, Graeme Laurie, Abbe Brown, Smita Kheria, and Jane Cornwell. *Contemporary intellectual property: law and policy*. Oxford University Press, USA, 2014.
- [5] Homaira Huda Shomee, Zhu Wang, Sathya N Ravi, and Sourav Medya. A survey on patent analysis: From nlp to multimodal ai. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8545–8561, 2025.
- [6] Yao Ding, Yuqing Wu, and Ziyang Ding. An automatic patent literature retrieval system based on llm-rag. *arXiv preprint arXiv:2508.14064*, 2025.
- [7] Homaira Huda Shomee, Suman Kalyan Maity, and Sourav Medya. PATENTWRITER: A benchmarking study for patent drafting with llms. *CoRR*, abs/2507.22387, 2025.
- [8] Dimitrios Christofidellis, Antonio Berrios Torres, Ashish Dave, Manuel Roveri, Kristin Schmidt, Sarath Swaminathan, Hans Vandierendonck, Dmitry Zubarev, and Matteo Manica. Pgt: a prompt based generative transformer for the patent domain. In *ICML 2022 workshop on knowledge retrieval and language models*, 2022.
- [9] Jieh-Sheng Lee. Instructpatentgpt: training patent language models to follow instructions with human feedback. *Artificial Intelligence and Law*, pages 1–44, 2024.
- [10] Juanyan Wang, Sai Krishna Reddy Mudhiganti, and Manali Sharma. Patentformer: a novel method to automate the generation of patent applications. In *Proceedings of the 2024 conference on empirical methods in natural language processing: industry track*, pages 1361–1380, 2024.
- [11] Zilong Bai, Ruiji Zhang, Linqing Chen, Qijun Cai, Yuan Zhong, Cong Wang, Yan Fang, Jie Fang, Jing Sun, Weikuan Wang, et al. Patentgpt: A large language model for intellectual property. *arXiv preprint arXiv:2404.18255*, 2024.
- [12] You Zuo, Kim Gerdes, Eric Villemonte de La Clergerie, and Benoît Sagot. Patenteval: Understanding errors in patent generation. In *NAACL2024-2024 Annual Conference of the North American Chapter of the Association for Computational Linguistics*, 2024.
- [13] Runtao Ren, Jian Ma, and Jianxi Luo. Large language model for patent concept generation. *Advanced Engineering Informatics*, 65:103301, 2025.
- [14] Yongmin Yoo, Qiongkai Xu, and Longbing Cao. Patentscore: Multi-dimensional evaluation of llm-generated patent claims. *arXiv preprint arXiv:2505.19345*, 2025.
- [15] Qiyao Wang, Shiwen Ni, Huaren Liu, Shule Lu, Guhong Chen, Xi Feng, Chi Wei, Qiang Qu, Hamid Alinejad-Rokny, Yuan Lin, et al. Autopatent: a multi-agent framework for automatic patent generation. *arXiv preprint arXiv:2412.09796*, 2024.

- [16] Valentin Knappich, Anna Häty, Simon Razniewski, and Annemarie Friedrich. PAP2PAT: benchmarking outline-guided long-text patent generation with patent-paper pairs. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar, editors, *Findings of the Association for Computational Linguistics, ACL 2025, Vienna, Austria, July 27 - August 1, 2025*, pages 9524–9554. Association for Computational Linguistics, 2025.
- [17] Xiang Liu, Peijie Dong, Xuming Hu, and Xiaowen Chu. Longgenbench: Long-context generation benchmark. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 865–883, 2024.
- [18] Yuhao Wu, Ming Shan Hee, Zhiqiang Hu, and Roy Ka-Wei Lee. Longgenbench: Benchmarking long-form generation in long context llms. In *International Conference on Learning Representations*, 2025.
- [19] Lekang Jiang and Stephan Goetz. Artificial intelligence exploring the patent field. *CoRR*, 2024.
- [20] Fiona Murray. Innovation as co-evolution of scientific and technological networks: exploring tissue engineering. *Research policy*, 31(8-9):1389–1403, 2002.
- [21] Tom Magerman, Bart Van Looy, and Xiaoyan Song. Exploring the feasibility and accuracy of latent semantic analysis based text mining techniques to detect similarity between patent documents and scientific publications. *Scientometrics*, 82(2):289–306, 2010.
- [22] Joshua S Gans, Fiona E Murray, and Scott Stern. Contracting over the disclosure of scientific knowledge: Intellectual property and academic publication. *Research Policy*, 46(4):820–835, 2017.
- [23] Fiona Murray and Scott Stern. Do formal intellectual property rights hinder the free flow of scientific knowledge? evidence from patent-paper pairs. *Northwestern University and MIT Sloan School: unpublished paper*, 2004.
- [24] Yiheng Xu, Minghao Li, Lei Cui, Shaohan Huang, Furu Wei, and Ming Zhou. Layoutlm: Pre-training of text and layout for document image understanding. In *Proceedings of the 26th ACM SIGKDD international conference on knowledge discovery & data mining*, pages 1192–1200, 2020.
- [25] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in neural information processing systems*, 33:9459–9474, 2020.
- [26] Iz Beltagy, Matthew E Peters, and Arman Cohan. Longformer: The long-document transformer. *arXiv preprint arXiv:2004.05150*, 2020.
- [27] Yunpeng Huang, Jingwei Xu, Junyu Lai, Zixu Jiang, Taolue Chen, Zenan Li, Yuan Yao, Xiaoxing Ma, Lijuan Yang, Hao Chen, et al. Advancing transformer architecture in long-context large language models: A comprehensive survey. *arXiv preprint arXiv:2311.12351*, 2023.
- [28] Marta R Costa-jussà, Pierre Andrews, Mariano Coria Meglioli, Joy Chen, Joe Chuang, David Dale, Christophe Ropers, Alexandre Mourachko, Eduardo Sánchez, Holger Schwenk, et al. Lcfo: Long context and long form output dataset and benchmarking. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 10672–10700, 2025.
- [29] Kimi Team, Yifan Bai, Yiping Bao, Guanduo Chen, Jiahao Chen, Ningxin Chen, Ruijue Chen, Yanru Chen, Yuankun Chen, Yutian Chen, et al. Kimi k2: Open agentic intelligence. *arXiv preprint arXiv:2507.20534*, 2025.
- [30] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jian Yang, Jiaxi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keqin Bao, Kexin Yang, Le Yu, Lianghao Deng, Mei Li, Mingfeng Xue, Mingze Li, Pei Zhang, Peng Wang, Qin Zhu, Rui Men, Ruize Gao, Shixuan Liu, Shuang Luo, Tianhao Li, Tianyi Tang, Wenbiao Yin, Xingzhang Ren, Xinyu Wang, Xinyu Zhang, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yinger Zhang, Yu Wan, Yuqiong Liu, Zekun Wang, Zeyu Cui, Zhenru Zhang, Zhipeng Zhou, and Zihan Qiu. Qwen3 technical report. *CoRR*, abs/2505.09388, 2025.
- [31] DeepSeek-AI. Deepseek-v3 technical report. *CoRR*, abs/2412.19437, 2024.
- [32] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.

- [33] Gheorghe Comanici, Eric Bieber, Mike Schaekermann, Ice Pasupat, Noveen Sachdeva, Inderjit Dhillon, Marcel Blistein, Ori Ram, Dan Zhang, Evan Rosen, et al. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. *arXiv preprint arXiv:2507.06261*, 2025.
- [34] Yushi Bai, Jiajie Zhang, Xin Lv, Linzhi Zheng, Siqi Zhu, Lei Hou, Yuxiao Dong, Jie Tang, and Juanzi Li. Longwriter: Unleashing 10,000+ word generation from long context llms. In *The Thirteenth International Conference on Learning Representations*, 2025.
- [35] Yuhao Wu, Yushi Bai, Zhiqiang Hu, Roy Ka-Wei Lee, and Juanzi Li. Longwriter-zero: Mastering ultra-long text generation via reinforcement learning. In *The Thirteenth International Conference on Learning Representations*, 2026.
- [36] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. React: Synergizing reasoning and acting in language models. In *The Eleventh International Conference on Learning Representations*, 2023.
- [37] Gaole He, Gianluca Demartini, and Ujwal Gadiraju. Plan-then-execute: An empirical study of user trust and team performance when using llm agents as a daily assistant. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, pages 1–22, 2025.
- [38] Barrett Lattimer, Patrick H Chen, Xinyuan Zhang, and Yi Yang. Fast and accurate factual inconsistency detection over long documents. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 1691–1703, 2023.
- [39] Chin-Yew Lin. Rouge: A package for automatic evaluation of summaries. In *Text summarization branches out*, pages 74–81, 2004.
- [40] Inez Okulska, Daria Stetsenko, Anna Kołos, Agnieszka Karlińska, Kinga Głabińska, and Adam Nowakowski. Stylometrix: An open-source multilingual tool for representing stylometric vectors. *arXiv preprint arXiv:2309.12810*, 2023.
- [41] Mauro Cettolo, Nicola Bertoldi, and Marcello Federico. The repetition rate of text as a predictor of the effectiveness of machine translation adaptation. In *Proceedings of the 11th Conference of the Association for Machine Translation in the Americas: MT Researchers Track*, pages 166–179, 2014.
- [42] Wei Zhao, Michael Strube, and Steffen Eger. Discoscore: Evaluating text generation with bert and discourse coherence. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 3865–3883, 2023.
- [43] Lekang Jiang and Stephan M Goetz. Natural language processing in the patent domain: a survey. *Artificial Intelligence Review*, 58(7):214, 2025.
- [44] Qiyao Wang, Jianguo Huang, Shule Lu, Yuan Lin, Kan Xu, Liang Yang, and Hongfei Lin. Ipeval: A bilingual intellectual property agency consultation evaluation benchmark for large language models. *CoRR*, abs/2406.12386, 2024.
- [45] Lekang Jiang, Caiqi Zhang, Pascal A. Scherz, and Stefan Goetz. Can large language models generate high-quality patent claims? In *Findings of the Association for Computational Linguistics: NAACL 2025, Albuquerque, New Mexico, USA, April 29 - May 4, 2025*, pages 1272–1287. Association for Computational Linguistics, 2025.
- [46] Shreya Shukla, Nakul Sharma, Manish Gupta, and Anand Mishra. Patentlmm: Large multimodal model for generating descriptions for patent figures. In *AAAI-25, Sponsored by the Association for the Advancement of Artificial Intelligence, February 25 - March 4, 2025, Philadelphia, PA, USA*, pages 20488–20496. AAAI Press, 2025.
- [47] Kevin Yang, Yuandong Tian, Nanyun Peng, and Dan Klein. Re3: Generating longer stories with recursive reprompting and revision. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 4393–4479, 2022.
- [48] Kevin Yang, Dan Klein, Nanyun Peng, and Yuandong Tian. Doc: Improving long story coherence with detailed outline control. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3378–3465, 2023.
- [49] Ishita Kumar, Snigdha Viswanathan, Sushrita Yerra, Alireza Salemi, Ryan A Rossi, Franck Dernoncourt, Hanieh Deilamsalehy, Xiang Chen, Ruiyi Zhang, Shubham Agarwal, et al. Longlamp: A benchmark for personalized long-form text generation. *arXiv preprint arXiv:2407.11016*, 2024.

- [50] Yiyan Xu, Jinghao Zhang, Alireza Salemi, Xinting Hu, Wenjie Wang, Fuli Feng, Hamed Zamani, Xiangnan He, and Tat-Seng Chua. Personalized generation in large model era: A survey. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 24607–24649, 2025.
- [51] Alireza Salemi, Cheng Li, Mingyang Zhang, Qiaozhu Mei, Weize Kong, Tao Chen, Zhuowan Li, Michael Bendersky, and Hamed Zamani. Reasoning-enhanced self-training for long-form personalized text generation. *arXiv preprint arXiv:2501.04167*, 2025.
- [52] Chau Minh Pham, Yapei Chang, and Mohit Iyyer. Clipper: Compression enables long-context synthetic data generation. *arXiv preprint arXiv:2502.14854*, 2025.
- [53] Chau Minh Pham, Simeng Sun, and Mohit Iyyer. Suri: Multi-constraint instruction following for long-form text generation. URL <https://arxiv.org/abs/2406.19371>, 2024.
- [54] Gang Chen, Changshuo Liu, Gene Anne Ooi, Marcus Tan, Zhongle Xie, Jianwei Yin, James Wei Luen Yip, Wenqiao Zhang, Jiaqi Zhu, and Beng Chin Ooi. Generative ai for healthcare: fundamentals, challenges, and perspectives. *Science China Information Sciences*, 69(7):173101, 2026.
- [55] Changshuo Liu, Wu Junran, Zhongle Xie, Wenqiao Zhang, Kaiping Zheng, Jiaqi Zhu, Qingpeng Cai, Ooi Gene Anne, Marcus Chun Jin Tan, Jianwei Yin, James Wei Luen Yip, and Beng Chin Ooi. DIYHealth suite: Dataset, model, and benchmark for health management at home. In *Forty-third International Conference on Machine Learning*, 2026.
- [56] Jiaqi Zhu, Shaofeng Cai, Fang Deng, Beng Chin Ooi, and Junran Wu. Do llms understand visual anomalies? uncovering llm’s capabilities in zero-shot anomaly detection. In *Proceedings of the 32nd ACM International Conference on Multimedia*, pages 48–57, 2024.
- [57] Mingjian Gao, Wenqiao Zhang, Yuqian Yuan, Yang Dai, Binhe Yu, Zheqi Lv, Haoyu Zheng, Jiaqi Zhu, Zhiqi Ge, Zixuan Wan, et al. Visualthink-vla: Visual intermediate reasoning for effective and low-latency vision-language-action policies. *arXiv preprint arXiv:2605.30011*, 2026.
- [58] Nan Zhou, Ke Zou, Meng Liu, Linchao He, Jiaqi Zhu, Yi Zhang, Hu Chen, and Huazhu Fu. Hallucination detection and correction in medical vlms via counter-evidence verification. *arXiv preprint arXiv:2606.18609*, 2026.
- [59] Nils Reimers and Iryna Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3982–3992, 2019.
- [60] Barbara J Grosz, Aravind Joshi, and Scott Weinstein. Centering: A framework for modeling the local coherence of discourse. *Computational linguistics*, 21(2):203–225, 1995.
- [61] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*, pages 4171–4186, 2019.
- [62] Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, et al. The llama 3 herd of models. *arXiv e-prints*, pages arXiv–2407, 2024.
- [63] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, Weizhu Chen, et al. Lora: Low-rank adaptation of large language models. *ICLR*, 1(2):3, 2022.
- [64] Qwen Team et al. Qwen2 technical report. *arXiv preprint arXiv:2407.10671*, 2(3), 2024.

Appendix

• A. Notation Table	15
• B. Extended Related Work	15
• C. Details on Semantic Chunking	16
• D. Workflow of Logic Tree Construction	16
• E. Patent Generation Prompt	18
• F. Implementation Details	20
· <i>F.1. Evaluation Metrics</i>	20
· <i>F.2. Baseline Methods</i>	21
· <i>F.3. Hyperparameters and Decoding Configurations</i>	22
· <i>F.4. Experimental Environment</i>	22
· <i>F.5. Detailed Prompt Settings</i>	22
• G. Supplementary Experimental Results	26
· <i>G.1. Supplementary Ablation Study</i>	26
· <i>G.2. Supplementary Study on Agentic Planning</i>	26
· <i>G.3. Evidence-support Analysis</i>	26
· <i>G.4. Computational Cost Analysis</i>	27
· <i>G.5. Human Expert Study</i>	27
· <i>G.6. Sensitivity to Tree Growth Budget</i>	28
· <i>G.7. Case Study</i>	29
• H. Discussion on Human Evaluation and Legal Adequacy	30

A Notation Table

To provide a comprehensive overview of the notations used throughout the paper, we present a summary of notations in Table 3 as a quick reference to facilitate the understanding and recall of each symbol.

Table 3: Notations.

Notation	Description
(X, P)	A patent-paper pair, where X is the source (pre-publication) research paper and P is the corresponding patent document.
$X = \{x_1, x_2, \dots, x_m\}$	Source paper represented as a collection of m text spans.
$P = \{p_1, \dots, p_n\}$	Patent document represented as a collection of n text spans.
$\tilde{X} = \{\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_L\}$	Semantic chunks produced from X after layout-aware semantic chunking, with $L \leq m$.
$\mathcal{T} = (V, E, r)$	Logic tree, a directed rooted tree with node set V , edge set $E \subseteq V \times V$, and root node r .
$V = \{v_1, \dots, v_N\}$	Set of logic tree nodes, each representing a technical element (e.g., principle, step, subsystem).
v_t	Current node dequeued for recursive generation at iteration t .
v_f	Parent node of v_t in the logic tree.
V_f^{child}	Set of child nodes of v_f .
$c(v)$	Generated text fragment associated with node v .
$\mathcal{E}(v)$	Supporting evidence spans from the source paper attributed to node v , $\mathcal{E}(v) \subseteq X$.
$\mathcal{V}_{kb}$	Vectorized document knowledge base built from semantic chunks for retrieval.
$\mathcal{E}_p(\cdot)$	Paragraph-level embedding function used for semantic similarity and indexing.
$\mathcal{M}_{LLM}$	Backbone large language model.
σ_I	Instruction prompt for generating an initial draft technical description.
$\mathcal{D}_d$	Draft technical description generated by $\mathcal{M}_{LLM}(\sigma_I, \tilde{X})$ to initialize the logic tree root.
$E(\cdot)$	Expansion procedure.
$R(\cdot)$	Refinement procedure.
σ_E	Expansion instruction prompt used to generate new child nodes from retrieved context.
σ_R	Refinement instruction set; in the draft $\sigma_R = \{\sigma_{R1}, \sigma_{R2}, \sigma_{R3}\}$.
$\mathcal{Q}$	FIFO queue used for breadth-first expansion over leaf nodes in recursive generation.
$\mathcal{D}_{ctx}$	Context evidence set used to expand or refine the current node v_t .
$S_{dr}(d)$	Discriminative relevance score for candidate span d .
τ_f	Threshold for retaining a candidate span via discriminative relevance.
τ_s	Semantic chunking threshold.
$t_{\max}$	Upper bound on accumulated token length for a chunk during semantic chunking.
T_d	A subtree allocated to the Detailed Description section.

B Extended Related Work

In this section, we provide an extended review of related work, encompassing both general long-text generation methods and patent-specific drafting paradigms.

B.1 Long-text Generation

Recent advances in long-text generation have explored structured prompting, hierarchical decomposition, and personalization to improve coherence over extended outputs [35]. Early approaches, such as recursive and outline-based generation [47, 48], introduce hierarchical planning to maintain narrative consistency across long contexts. More recent work further incorporates personalization [49, 50] and reasoning-oriented training [51], enabling models to better align outputs with user intent.

With growing demand for ultra-long generation (e.g., beyond 2k tokens), new datasets and training paradigms have emerged. For instance, large-scale instruction-following datasets constructed via back-translation support long-context learning but often rely on synthetic supervision and remain limited in output length [52, 53]. Other approaches, such as LongWriter [34], leverage supervised and preference-based optimization on agent-generated long-form data to extend output length, but may inherit biases from teacher models and lack explicit control over global structure. LongWriter-Zero [35] further explores reinforcement learning from scratch to induce ultra-long generation without relying on synthetic supervision, improving length control and structural consistency. Overall, despite notable progress, existing approaches lack explicit global structure modeling and fine-grained evidence grounding, limiting their ability to ensure coherent, factual, and controllable generation over ultra-long contexts.

Beyond long-form generation, domain-specific LLM systems have explored complementary mechanisms for evidence organization, contextual reasoning, and output verification. Data-centric generative systems can organize heterogeneous knowledge through semantic vector search, contextual querying, and agentic retrieval backends [54, 55]. Fine-grained evidence alignment and verification have also been studied through global-to-local semantic alignment [56], compact visual-evidence interfaces for intermediate reasoning [57], and claim-level counter-evidence verification for hallucination detection and correction [58]. These studies

collectively illustrate the broader value of explicitly organizing and validating evidence during model reasoning and generation.

B.2 Patent Drafting

While prior work has studied long-form generation in general domains, patent drafting introduces additional challenges due to strict structural, legal, and evidence-grounding requirements. We next review patent drafting, covering both benchmark development and generation paradigms.

Patent-related Benchmarks. With the advancements of LLMs, the effort of testing their abilities in patent-related tasks has emerged. Recent progress in LLMs has spurred benchmarking of patent-centric capabilities [43]. For example, IPEval [44] tests LLMs for patent law QA. PatentEval [12] focuses on the tasks of claims-to-abstract and next-claim generations. HUPD-DCG [45] measures patent claim generation performance from patent descriptions. Complementary to task-specific benchmarks, Pap2Pat [16] extends evaluation to long-text patent drafting, reflecting growing interest in document-level generation. These benchmarks illustrate the evolution of patent-related evaluation from localized capabilities toward holistic long-form technical writing.

Patent-related Generation. Existing patent generation methods can be broadly grouped into three paradigms. First is *patent-to-patent*, where a section of the patent content is generated from those of specific parts in the same patent. For instance, PGT [8] is pre-trained to support multiple patent-related tasks, e.g., title-to-abstract, abstract-to-claim. PatentLMM [46] presents a large multimodal model to write descriptions of patent figures. [13] proposes a knowledge-tuning approach for initial concept discovery and content generation. Second is *draft-to-patent*, where a draft of a patent is expanded and aligned with patent format and wording. PatentGPT [11] pre-trains a series of base LLMs and benchmarks them with patent generation from technical disclosure documents draft. AutoPatent [15] generates patents from draft information, where the draft is constructed through multiple rounds of iterative question–answering with an LLM under the guidance of a reference patent document. Third is *paper-to-patent*, where an end-to-end process is studied from the original research paper to the patent draft. For example, Pap2Pat [16] leverages ground-truth patent-derived outlines to guide the full-text patent migration. However, prior methods either operate under partial-context settings or depend on externally provided (often manually crafted) drafting priors, leaving realistic long-form patent drafting with global logical organization less well addressed.

C Details on Semantic Chunking

To convert the raw content of the input document into knowledge units suitable for retrieval-augmented generation, we first perform semantic structure reconstruction based on page-level metadata extracted by PyMuPDF¹, including textual blocks, font styles, coordinates, line spans, and paragraph layouts. Using these layout cues, the system identifies the hierarchical boundaries of sections and paragraphs, thereby reconstructing the document’s logical structure.

Building upon this structure, we conduct semantic chunking to produce contextually coherent text segments that serve as the foundational elements of the vectorized document knowledge base. Unlike fixed-length segmentation methods [27, 16], our approach dynamically determines chunk boundaries according to the semantic coherence and contextual dependency between adjacent paragraphs. Formally, given a document represented as an ordered sequence of paragraphs $\tilde{X} = \{\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_m\}$, we compute the pairwise semantic similarity $s_{i,i+1} = \text{sim}(\mathcal{E}_p(\tilde{x}_i), \mathcal{E}_p(\tilde{x}_{i+1}))$, where $i = 1, \dots, m-1$, and $\mathcal{E}_p(\cdot)$ denotes the paragraph-level embedding function (e.g., Sentence-BERT [59]). The pairwise semantic similarity $\text{sim}(\cdot, \cdot)$ can be instantiated with different similarity metrics, while cosine similarity is adopted in this work due to its efficiency and empirical stability. Paragraphs are iteratively merged when $s_{i,i+1} \geq \tau_s$ and the accumulated token length does not exceed the upper bound t_{max} . This procedure yields a set of semantically coherent chunks $X = \{x_1, x_2, \dots, x_L\}$, where $L \leq m$ represents the total number of chunks for the input document and x_i represents the i -th chunk aligned with the document’s logical hierarchy. The resulting chunks constitute the minimal retrieval units for RAG, which are embedded into a high-dimensional semantic space and stored in the vector database as a vectorized document knowledge base, supporting structured querying, traceable grounding, and multi-turn semantic reasoning.

D Workflow of Logic Tree Construction

In this section, we provide the detailed algorithmic specification of the logic tree construction process used in LogicTree-RAG. As described in Section 3.5, logic tree construction constitutes the first stage of the overall patent generation pipeline, aiming to transform an unstructured research paper into a structured, evidence-grounded hierarchical representation. The procedure integrates semantic chunking, document-level vectorization,

¹<https://pymupdf.readthedocs.io/>

E Patent Generation Prompt

LogicTree-RAG employs a structured set of prompt instructions to control different stages of logic tree construction and recursive patent generation. Specifically, we design three categories of prompts: an initialization instruction σ_I , an expansion instruction σ_E , and a refinement instruction σ_R . These prompts act as a lightweight control interface for the backbone LLM, enabling evidence-grounded, controllable long-form patent drafting.

The initialization instruction σ_I is used to generate a draft technical description $\mathcal{D}_d$, which serves as the root node of the logic tree. Given a set of semantically coherent chunks extracted exclusively from the introduction of the source paper, σ_I prompts the model to synthesize a high-level technical description that captures the problem setting, core solution, and key system or method components, as detailed below.

```

1  ### ROLE
2  You are an expert patent drafter and technical editor.
3
4  ### TASK DESCRIPTION
5  Your task is to write a draft technical description of the invention, based ONLY on the provided
   document chunks of a research paper.
6
7  ### INPUT
8  Input: a set of semantically coherent chunks derived exclusively from the paper introduction.
9
10 Document chunks:
11 """
12 {CHUNKS}
13 """
14
15 ### REQUIREMENTS
16 Write a draft technical description that satisfies ALL requirements:
17 1. Capture the invention at a system and method level: clearly state (i) the technical problem, (ii) the
   core technical solution, and (iii) the key technical components.
18 2. Make the description expansion-ready: include explicit technical handles such as module names/roles,
   input-output signals, key variables/parameters, and the end-to-end workflow, so that each part can
   be expanded later into detailed patent sections.
19 3. Do not cite the paper, authors, or section numbers. Do not mention "chunks" or "retrieval".
20 4. Use ONLY information supported by the chunks.
21 5. No numbering, no bullet points, no headings.
22 6. Avoid vague hedges such as "may", "could", or "for example". Prefer definitive, technical phrasing.
23
24 Output ONLY the draft technical description text, with no extra commentary.

```

Listing 1: Refinement instruction σ_{R1} for under-specified term detection.

The expansion instruction σ_E governs logic tree growth. Conditioned on a node contribution and its associated context evidence, σ_E prompts the model to elaborate the contribution into finer-grained algorithmic details. The expansion instruction σ_E is specified as follows.

```

1  ### ROLE
2  You are an expert in polishing patent applications, with professional knowledge of patent drafting and
   formatting standards.
3
4  ### TASK DESCRIPTION
5  Your task is: given a specific claim contribution, extract the concrete algorithmic steps from the
   provided context and use them to expand the contribution, further specifying its algorithmic
   details.
6
7  ### REQUIREMENTS
8  1. The output should consist of several paragraphs without numbering. The first paragraph must restate
   the given contribution point "{text}", and the subsequent paragraphs should describe the detailed
   algorithmic steps. Separate each paragraph with a blank line.
9  2. The expansion must remain strictly focused on the content and concept described by the given
   contribution point, without introducing broader or unrelated concepts.
10 3. The total output length should be longer than the input.
11 4. When necessary, include key formulas and definitions. Do not insert line breaks inside formulas.
12 5. Maintain precision and conciseness. Avoid terms such as "for example" or "may".
13 6. Each paragraph must be sufficiently detailed. If a step involves multiple sub-steps, describe them
   within a single paragraph using multiple sentences. The granularity of the steps should be
   consistent, with each step being necessary and indispensable.
14
15 The original contribution point is: {TEXT}\n\n
16 The provided context is: {CONTEXT}\n\n

```

Listing 2: Expansion instruction σ_E for LogicTree-RAG in long-form patent drafting.

The refinement instruction set $\sigma_R = \{\sigma_{R1}, \sigma_{R2}, \sigma_{R3}\}$ addresses under-specified content during recursive generation. Specifically, σ_{R1} detects under-specified terms that lack sufficient technical detail, σ_{R2} refines such terms using retrieved supporting evidence when available, and σ_{R3} completes refinement using the model's internal knowledge when no relevant evidence exists. The refinement instructions σ_R are instantiated as follows.

```

1  ### ROLE
2  You are an expert patent drafter and technical editor.
3
4  ### TASK DESCRIPTION
5  Your task is to identify under-specified terms in a patent specification node and propose how they
   should be refined.
6
7  ### DEFINITION
8  Definition (under-specified term):
9  A term/phrase is under-specified if, for a technical disclosure, it lacks at least one of the following:
   a clear definition, a concrete algorithmic procedure, required inputs/outputs, parameters/
   hyperparameters, constraints/assumptions, or implementation-level details.
10
11 ### INPUT
12 Input node content:
13 """
14 {NODE_CONTENT}
15 """
16
17 ### REQUIREMENTS
18 1. Each "term" must be an exact substring of the input.
19 2. Do not rewrite the node content. Do not add any extra keys.
20 3. Prefer terms that are essential for technical completeness (e.g., key components, variables,
   algorithms, objectives, constraints).
21 4. Output ONLY the terms, one per line, with no numbering, no bullets, no extra text.

```

Listing 3: Refinement instruction σ_{R1} for under-specified term detection.

```

1  ### ROLE
2  You are an expert patent drafter and technical editor. You will refine a specific under-specified term
   in a patent-specification node using the provided evidence.
3
4  ### TASK DESCRIPTION
5  Your task is to produce an updated version of the node content that makes the target term "{TERM}" fully
   specified in a patent-appropriate and technically reproducible way.
6
7  ### INPUT
8  Target term to refine:
9  "{TERM}"
10
11 Original node content:
12 """
13 {NODE_CONTENT}
14 """
15
16 Retrieved evidence (context spans):
17 """
18 {EVIDENCE_SPANS}
19 """
20
21 ### REQUIREMENTS
22 1. All newly added technical details MUST be supported by the evidence.
23 2. Preserve the original wording and structure as much as possible. Only add or revise the minimal
   necessary text to specify "{TERM}".
24 3. When applicable, specify inputs/outputs, key variables/parameters, computation steps, constraints/
   assumptions, and any formulas explicitly stated or implied by the evidence.
25 4. Use formal, precise language. Avoid hedges such as "may", "could", "for example". Avoid
   conversational tone.
26
27 Refined node content:

```

Listing 4: Refinement instruction σ_{R2} for evidence-grounded targeted refinement.

```

1  ### ROLE
2  You are an expert patent drafter and technical editor. You will refine an under-specified term in a
   patent-specification node.
3
4  ### TASK DESCRIPTION
5  Your task is to update the node content to make "{TERM}" sufficiently specified for a patent
   specification, using general and widely applicable technical descriptions.
6
7  ### INPUT

```

```

8 Target term to refine:
9 "{TERM}"
10
11 Original node content:
12 ""
13 {NODE_CONTENT}
14 ""
15
16 ### REQUIREMENTS
17 1. Preserve the original wording and structure as much as possible. Only add or revise the minimal
   necessary text to specify "{TERM}".
18 2. Provide a standard, implementation-oriented specification that would be reasonable across typical
   systems (e.g., define inputs/outputs, typical parameterization, and a generic procedure).
19 3. Use formal, precise language. Avoid hedges such as "may", "could", "for example". Avoid
   conversational tone.
20
21 Refined node content:

```

Listing 5: Refinement instruction σ_{R3} for knowledge completion

F Implementation Details

F.1 Evaluation Metrics

We evaluate patent generation performance from both content-level and language-level perspectives. Given the length and specialized nature of patent documents, automatic evaluation requires disentangling semantic content quality from linguistic realization. We therefore adopt established metrics that are well-suited for long-form technical text generation.

Content-level Metrics. Content-level metrics evaluate whether the generated patent conveys accurate and sufficient technical information grounded in the reference materials. All content-level metrics are computed at the document level following the evaluation protocol [16].

- **Coverage.** Coverage measures the extent to which the generated patent covers the semantic content of the reference patent. We adopt the NLI-based SCALE metric [38], which estimates entailment between two long documents by aggregating sentence-level entailment scores.

Given a generated patent X and a reference patent R , we treat X as the *premise document* and R as the *hypothesis document*. Due to the quadratic cost of exhaustive sentence-level comparisons, SCALE samples a set of hypothesis sentences $\{h_j\}_{j=1}^N$ from R . For each hypothesis sentence h_j , the top- K most relevant premise chunks $\{p_{j,k}\}_{k=1}^K$ are retrieved from X using BM25. An NLI model then computes entailment probabilities $P_{\text{entail}}(p_{j,k}, h_j)$. The sentence-level entailment score is defined as:

$$s(h_j) = \max_k P_{\text{entail}}(p_{j,k}, h_j). \quad (7)$$

The coverage score is computed by averaging over all sampled hypothesis sentences:

$$\text{Coverage}(X, R) = \frac{1}{N} \sum_{j=1}^N s(h_j). \quad (8)$$

- **Patent-grounded Factuality.** Patent-grounded factuality ($\mathcal{F}_{Pat}$) measures the extent to which the generated patent is supported by the reference patent alone. In this setting, the reference patent R serves as the premise document and the generated patent X serves as the hypothesis document:

$$\mathcal{F}_{Pat}(X, R) = \text{SCALE}(R \Rightarrow G). \quad (9)$$

- **Source-grounded Factuality.** Source-grounded factuality ($\mathcal{F}_{Src}$) measures semantic consistency when both the reference patent and the source paper are used as grounding evidence. Specifically, the premise document is formed by concatenating the reference patent R and the source paper P , while the generated patent X remains the hypothesis document:

$$\mathcal{F}_{Src}(X, R, P) = \text{SCALE}(R \cup P \Rightarrow G). \quad (10)$$

This variant accounts for technically correct information present in the source paper but absent from the granted patent.

- **ROUGE-L.** ROUGE-L evaluates the longest common subsequence (LCS) between predicted and ground truth answers, reflecting recall-oriented similarity. It captures global structural overlap and complements BLEU’s precision-based evaluation.

$$\text{ROUGE-L} = \frac{(1 + \beta^2) \cdot R_{lcs} \cdot P_{lcs}}{R_{lcs} + \beta^2 \cdot P_{lcs}}, \quad (11)$$

where $R_{lcs} = \frac{\text{LCS}(X,Y)}{|Y|}$, $P_{lcs} = \frac{\text{LCS}(X,Y)}{|X|}$. Y is the ground truth patent, i.e., the reference patent, X is the predicted text, i.e., the generated patent. $\text{LCS}(X, Y)$ is the length of the longest common subsequence. β controls the weighting of recall versus precision.

Language-level Metrics. Language-level metrics evaluate whether the generated patent adheres to stylistic conventions of patent writing, maintains discourse coherence, and avoids degenerate generation behaviors.

- **Style.** We measure stylistic similarity using corpus-wide n -gram profiles and StyloMetrix [40], as in [16]. For n -gram profiles, we construct frequency distributions over the 1,000 most frequent n -grams for $n \in \{1, 2, 3, 4\}$ from the reference patents and the generated patents. Let $P_{\text{ref}}^n(g)$ and $P_{\text{gen}}^n(g)$ denote the normalized frequencies of n -gram g . The similarity score for each n is computed as:

$$\text{sim}_n = \sum_{g \in P_{\text{ref}}^n \cup P_{\text{gen}}^n} \left(1 - \left(\frac{P_{\text{ref}}^n(g) - P_{\text{gen}}^n(g)}{P_{\text{ref}}^n(g) + P_{\text{gen}}^n(g)} \right)^2 \right). \quad (12)$$

We additionally compute a StyloMetrix similarity score based on 196 rule-based linguistic features, including verb tense, modality, POS distributions, lexical patterns, and syntactic constructions. The final style score is computed as:

$$\text{Style} = \frac{1}{5} \left(\sum_{n=1}^4 \text{sim}_n + \text{sim}_{\text{stylo}} \right). \quad (13)$$

- **Repetition Rate.** Repetition Rate ($\mathcal{R}_r$) is defined as the fraction of n -grams that appear more than once within a text [41], given by:

$$\mathcal{R}_r = \frac{\#\{\text{n-grams occurring more than once}\}}{\#\{\text{total n-grams}\}}. \quad (14)$$

To capture long-range repetition and degeneration effects, $\mathcal{R}_r$ is computed over sliding windows of 256 tokens and averaged across windows.

- **Coherence.** Discourse-level coherence is evaluated using DiscoScore [42], a coherence metric based on Centering Theory [60] and contextualized representations from BERT [61]. DiscoScore assesses the consistency of entity transitions and discourse structure across adjacent sentences, producing a document-level coherence score. Given a hypothesis document X , let $\mathbf{S}_{\text{gen}} \in \mathbb{R}^{n \times d}$ denote the matrix of sentence embeddings, where n is the number of sentences and d is the embedding dimension. A sentence graph $\mathcal{G}_{\text{gen}} = (V, A_{\text{gen}})$ is constructed, where each node corresponds to a sentence and the adjacency matrix A_{gen} encodes discourse connectivity between sentence pairs. Sentence embeddings are updated via graph propagation $\hat{\mathbf{S}}_{\text{gen}} = (A_{\text{gen}} + I)\mathbf{S}_{\text{gen}}$, where I is the identity matrix.

The same procedure is applied to the reference document R to obtain $\hat{\mathbf{S}}_{\text{ref}}$. A document-level representation is then computed by aggregating sentence embeddings, yielding $\mathbf{G}_{\text{gen}}$ and $\mathbf{G}_{\text{ref}}$. The final DiscoScore is defined as the cosine similarity between the two graph-level representations:

$$\text{DiscoScore}(X, R) = \cos(\mathbf{G}_{\text{gen}}, \mathbf{G}_{\text{ref}}). \quad (15)$$

Together, these metrics offer a comprehensive evaluation of patent generation quality that captures content accuracy, linguistic conventions, discourse coherence, and long-form generation behavior.

F.2 Baseline Methods

For heuristic baselines, including the ground-truth patent and source paper, no additional model inference is performed, and the documents are directly treated as generated outputs for metric computation.

Single LLM-call baselines generate the entire patent document in a single LLM invocation by conditioning on the full input paper using long-context prompting. All baselines support 128k-token contexts, enabling full-paper ingestion and long-text drafting. No intermediate structure or iterative refinement is employed. Specifically, Kimi K2 [29] adopts a Mixture-of-Experts (MoE) architecture with 1T total parameters and 32B activated per token, and is pre-trained on 15.5T tokens with carefully engineered optimization strategies to ensure stable training at scale. Qwen3-80B [30] follows the Qwen3-Next design and integrates hybrid attention mechanisms and high-sparsity MoE layers, enabling efficient context modeling under ultra-long context lengths while maintaining computational efficiency. Qwen3-Max [30] is a proprietary flagship model exceeding one trillion parameters, pre-trained on 36T tokens, and inherits the MoE-based design of the Qwen3 series with enhanced load balancing and stability-oriented training strategies. DeepseekV3 [31] is an MoE model with 671B total parameters and 37B activated per token, combining Multi-head Latent Attention and DeepSeekMoE architectures, and is pre-trained on 14.8T tokens with multi-token prediction objectives to support efficient and stable large-scale training. GPT-5 [32] and Gemini-2.5 Pro [33] are proprietary frontier LLMs evaluated under deep thinking mode, which enables stronger deliberative reasoning and more careful long-context synthesis.

They serve as strong closed-source baselines for assessing whether advanced reasoning-oriented prompting alone can produce coherent and complete long-form patent drafts without explicit structural orchestration. Collectively, these models provide strong capacity for long-context representation and serve as competitive backbone LLMs for single LLM-call patent generation. Unless otherwise specified, these baselines use the same decoding configurations as LogicTree-RAG.

Furthermore, we include Chunk-based Outline-guided Patent Generation (COPGEN) [16] and its supervised fine-tuned (SFT) variant as representative multi-stage LLM-based patent generation methods. Specifically, COPGEN chunks the outline, selects the most relevant parts of the source paper depending on the chunks’ outline bullet points, and prompts an LLM to generate the patent text for that chunk. To ensure a fair comparison under our problem setting, where no ground-truth patent outline is available at inference time, we report two variants of COPGEN. The first follows the outline-empty configuration, where only section titles are provided. For completeness, we also report results obtained using the long-version outlines provided by the Pap2Pat benchmark. For the SFT variant, only the backbone LLM is fine-tuned on the Pap2Pat training set (1,000 paper-patent pairs), and no test data is used during training.

In addition, we consider recent long-form generation frameworks LongWriter [34] and LongWriter-Zero [35], which decompose long-form generation into multi-stage processes to improve global coherence and content planning. We also include general-purpose agentic paradigms, including ReAct [36] and Plan-then-Execute [37], which enable LLMs to perform step-wise planning and tool use without relying on hard-coded structural templates. To adapt them to our setting, we expose our core components, i.e., semantic chunking, node-aware retrieval, and evidence-guided generation, as callable tools, and replace the logic-tree-based orchestration with generic agent planning strategies.

F.3 Hyperparameters and Decoding Configurations

In this subsection, we summarize the hyperparameters and decoding configurations adopted for LogicTree-RAG and the baseline methods. LogicTree-RAG adopts Qwen3-80B [30] as the backbone LLM throughout all experiments, unless stated otherwise. We set the semantic chunking threshold τ_s to 0.8, and the accumulated token length t_{max} to 512. For node-aware semantic search, we first retrieve up to 100 candidate evidence spans $\mathcal{D}_{ctx}$ from the document knowledge base $\mathcal{V}_{kb}$. These candidates are then re-ranked using the discriminative relevance score. Rather than using a fixed score threshold, we retain the top-5 candidates and set the discriminative relevance threshold τ_f adaptively as the score of the 5-th ranked candidate. To ensure controllable expansion during evidence-guided recursive generation, we impose a content length budget B that limits the maximum size of the logic tree. Specifically, the recursive process terminates when the number of nodes in the tree reaches 64, preventing unbounded expansion while preserving sufficient structural coverage. In addition, we constrain the maximum depth of the logic tree to 3 (with the root node at depth 0), which avoids overly deep recursive expansion and encourages a balanced hierarchical structure. Unless otherwise specified, these values are fixed across all experiments.

Reported results in Section 4.2 are evaluated on the Pap2Pat test set, which consists of 500 paper-patent pairs. For the SFT variant of COPGEN, fine-tuning is conducted exclusively on the Pap2Pat training set containing 1,000 pairs. All LLM-based methods use identical decoding configurations, with a temperature of 0.3, top- p of 1.0, and a maximum generation length of 8,192 tokens. The SFT variant of COPGEN fine-tunes the backbone LLM Llama-3 8B [62] using LoRA [63]. Following [16], we adopt a maximum sequence length of 8,192 tokens and train the model for 3 epochs with a batch size of 32. The learning rate is set to 3.16×10^{-4} with a cosine scheduler and a warmup ratio of 0.1. For LoRA-specific configurations, we use a rank of 128, a scaling factor (α) of 60, and a dropout rate of 0.05.

F.4 Experimental Environment

All experiments are conducted on a server equipped with an AMD EPYC 9334 CPU @ 2.7 GHz (32 cores), 128 GB of memory, and an NVIDIA H100 NVL with CUDA 12.9. The operating system is Ubuntu 24.04 running Linux kernel 6.8.0-79-generic. LLMs used in our experiments are accessed via standardized cloud-based inference APIs provided by Alibaba Cloud Bailian.

F.5 Detailed Prompt Settings

This subsection details the prompt configurations used in our experiments. For COPGEN and its variants, we adopt the prompt settings from the publicly released implementation². For single LLM-call baselines, we employ a unified prompt template following COPGEN to ensure a fair and consistent comparison, as described below.

1 ### ROLE

²<https://github.com/boschresearch/Pap2Pat>

```

2 You are a highly skilled patent attorney with decades of experience in drafting high-quality patent
  applications.
3 You assist scientists in transforming their scientific discoveries into lucrative patents.
4
5 ### TASK DESCRIPTION
6 Your task is to draft a patent application.
7
8 ### INPUTS
9 As input, you will be provided a research paper and a patent outline, each serving a distinct purpose.
10
11 1. Research Paper:
12 The research paper describes a novel invention to be patented. Your task is to extract the invention
  from the paper and write a patent application for it.
13
14 2. Patent Outline:
15 The patent outline summarizes the desired discourse structure of the patent document. Use this outline
  as a rough guidance during drafting.
16
17 ### GUIDELINES
18 * Copy the headings from the outline exactly. You must include only the headings provided in the outline.
19
20 * You must always write complete sentences and avoid keywords, bullet lists and enumerations.
  * The patent must act as a standalone document, therefore do not refer to the research paper in the
  patent.

```

Listing 6: System prompt for single LLM-call baselines in long-form patent drafting.

```

1 Here is the outline of the desired patent application.
2 Per bullet point, write roughly 4000 words.
3
4 ### DESCRIPTION OF THE INVENTION
5 In the example above, each line beginning with '#' is a bullet point.
6 '''md
7 # DESCRIPTION
8
9 ## FIELD OF THE INVENTION
10
11 ## BACKGROUND OF THE INVENTION
12
13 ## SUMMARY OF THE INVENTION
14
15 ## DETAILED DESCRIPTION
16 '''
17 You need to draft a complete patent application that strictly follows the outline's section order and
18 headings.
19 Do not skip any bullet points. Use formal patent language. Remember what you have already written and do
20 not repeat yourself.
21
22 ### RESEARCH PAPER
23 Here is the research paper that describes the invention:
24 '''
25 # Introduction
26
27 Safe delivery of a highly conformal dose distribution to a well-defined target volume in radiotherapy is
    not an easy task. It has become increasingly challenging due to the advent of advanced treatment
    techniques such as intensity-modulated radiation therapy (IMRT) and volumetric-modulated arc
    therapy (VMAT). Though a downward trend in radiotherapy incident rates has been indicated by
    several reports (1, 2), severe incidents with detrimental effects, including death, have been
    reported recently and received public attention. (3) Radiotherapy is a complicated, multistep,
    multiperson process, and errors can occur at any point. One of the prominent causes for
    radiotherapy incidents is geometric miss due to incorrect patient setup, which leads to the
    treatment of incorrect body parts with more than 1 cm spatial discrepancy (3, 4). Geometric miss
    can result in significant underdose to the target, which can cause tumor recurrence, and overdose
    to healthy tissue with severe normal tissue complications. In hypofractionated radiotherapy, such
    as stereotactic body radiotherapy (SBRT), it may result in even more severe morbidity than
    traditional radiotherapy.
28
29 Several reports have identified geometric miss caused by incorrect patient setup as the leading cause of
    radiotherapy errors. In a study of 100 radiotherapy incidents reported internationally, (3) the
    frequency of incidents due to incorrect patient setup was reported to be 21%. Excluding the 44
    incidents involving brachytherapy, in which patient setup accuracy was less of an issue, this
    frequency would rise significantly to 37.5%. (3) Clark et al. (2) analyzed clinical incidents
    reported internally within a large academic center between 2007 and 2009, and reported that 14 out
    of 41 critical, major, or serious incidents were geographic misses. Among the 14 incidents, one was
    caused by wrong target identification in treatment planning, while the other 13 incidents were due
    to shifting errors at patient setup. In an online report analyzing event causes for 230
    misadministrations reported between 2001 and 2009 in New York from the New York State Department of
    Health, incorrect body part treated was the leading type of errors at 46%, most often due to
    incorrect patient setup. (5)
30
31 The challenge in patient setup is to accurately localize the patient to the same treatment position as
32 planned in each treatment session. Over the past decade, there has been a rapid expansion in
    technological tools to facilitate accurate patient localization. Examples include immobilization
    devices with couch indexing capability and image-guided radiotherapy (IGRT). Indexed immobilization
    devices not only ensure repeatable patient fixation but also reduce patient setup errors by
    providing initial approximate target localization. X-ray-based image-guided technologies, such as
    cone-beam CT (CBCT), enable visualization of internal anatomy with sufficient soft tissue contrast
    and allow corrections for misalignment or interfraction motion through registration with reference
    CT images (6). IGRT systems capable of continuous tracking, such as AlignRT (Vision RT Ltd., London,
    UK), C-Rad Sentinel (C-Rad AB, Uppsala, Sweden), SonArray (Varian Medical Systems, Palo Alto, CA),
    and ExacTrac Optical Tracking System (BrainLAB, Heimstetten, Germany), have also been developed
    for patient setup guidance. These systems have the potential to eliminate patient setup errors and
    significantly reduce setup uncertainty, as demonstrated by multiple studies. Bissonnette and Medlam
    (1) reported a 50 percent overall decrease in incidents caused by localization errors following
    the widespread introduction of IGRT on six of their 16 linear accelerators. Clark et al. (2) noted
    that none of the 13 geometric misses occurred on treatment units equipped with daily image guidance
    systems.
30
31 However, the use of advanced technological equipment does not guarantee that radiotherapy is immune to
32 setup errors (7, 8). Patient setup is a process that cannot be fully automated and is therefore
    subject to human error. Several contributing factors have been identified. First, sufficient formal
    training is not always provided to personnel who operate the devices and interpret the results.
    Incorrect interpretation can lead to improper adjustment of treatment position and consequently to
    setup errors. For example, when treating the thoracic spine, there is a significant risk of
    targeting the wrong vertebral body (9). Due to similarities in bony anatomy in this region,
    incorrect alignment may occur as a result of misregistration using orthogonal imaging or CBCT.

```

Second, as the complexity of treatment tools increases, the complexity of device control and the workload for therapists also increase, as reflected by the growing number of computer monitors in control rooms (8). In the absence of streamlined workflows and standardized controls, therapists may lose attention to the correctness of treatment delivery. Third, although most IGRT devices provide high geometric precision (10), they have inherent limitations that may compromise patient safety. For example, commonly used radiographic systems such as CBCT do not track patient position changes and therefore represent patient position only at the time of image acquisition. After CBCT imaging, the treatment couch may be moved for specific reasons, such as clearance checks, and not returned to the intended treatment position. Modern delivery systems include interlocks to detect such errors but also allow user overrides. If an interlock is inappropriately overridden, the patient may be treated at an incorrect site. In addition, CBCT and most radiographic-based IGRT systems cannot be used for noncoplanar setups, resulting in discrepancies between imaging and treatment positions when noncoplanar beams are applied (11). Continuous tracking systems such as ExacTrac Optical Tracking System, AlignRT, and C-Rad Sentinel are typically employed only for limited disease sites in current clinical practice. Therapists operating machines equipped with multiple IGRT systems often switch between devices depending on the treatment site, further increasing operational complexity. Fourth and most importantly, immediate and independent position verification is not always available to confirm that the patient has been positioned exactly as planned, particularly when the limitations of the employed IGRT systems are encountered (8).

In this paper, we present an efficient automatic patient safety system (PSS) that addresses factors contributing to gross setup errors and safeguards patient treatment. The system uses CCD cameras to track a single infrared reflective marker affixed to the patient's skin or immobilization device. It is a general-purpose system applicable to all disease sites and provides continuous and independent verification of patient setup accuracy. With a fully automated workflow, the PSS serves as an effective complement to existing IGRT systems to enhance patient safety in radiotherapy.

Materials And Methods

A. System Overview

The system consists of a pair of Polaris CCD cameras (Northern Digital Inc., Waterloo, Ontario) mounted on the treatment room ceiling (see Fig. 1(a)), a single IRRM affixed to the patient's skin or immobilization device, and a set of in-house developed software. The software communicates with the control box of the cameras through a standard serial cable and a proprietary data cable. An internal processing unit performs triangulation to determine the coordinates of IRRMs in its native coordinate system. The three-dimensional (3D) coordinates of up to 50 IRRMs can be transmitted to the computer simultaneously. System calibration, workflow, and daily quality assurance (QA) procedures are described in the following sections.

B. System Calibration

A calibration procedure was developed to convert the camera's native coordinate system to the absolute room coordinate system. A calibration jig consisting of five commercial IRRMs (Fig. 1(b)) was used for this purpose. The coordinates of each IRRM relative to the center of the jig were determined via CT scan and collected in a 5x3 matrix A. Under the guidance of CBCT, the jig was placed on the treatment couch with its center aligned with the imaging isocenter. The congruence of the imaging isocenter and the machine isocenter was confirmed using a ball bearing phantom (6). The coordinates of the IRRMs were sampled by the cameras and collected into another matrix B. A rotation matrix R and a translation matrix S were estimated by minimizing the residual error between the transformed and reference coordinates.

This is known as the relative pose problem in computer vision and has been studied extensively. 12, 13, 14 In this work, we employed a general analytical solution via singular value decomposition (SVD), (14) and the steps were summarized as follows: The centroids (A_c) and (B_c) were subtracted from A and B, respectively, to align the origins of the two coordinate systems. The SVD of $BT A$ was computed as $(B^T A) = U \{UV^T\}$ where U and V are unitary matrices, and W is a diagonal matrix whose diagonal entries are equal to the singular values of $BT A$.

3. The solutions of R and S were given as
$$\begin{matrix} R = U \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \det(UV^T) \end{pmatrix} V^T \\ S = A_c - B_c R \end{matrix}$$

System Workflow

Great emphasis was placed on developing a smooth clinical workflow for the system. The workload added to the therapists should be kept to a minimum, which was critical for the system to be adopted into clinic use. A minimal workload would also help minimize the possibilities of human errors. The workflow of the PSS is described in the following sections as the preparation stage and the treatment stage. Figure 2 shows a demonstrative workflow of the system when used on head and neck (H&N) patients treated with aquaplastic face masks.

...

Conclusions

We have developed an efficient automatic general-purpose PSS to prevent gross setup errors in radiotherapy. The system provides real-time independent position verification for the treatment of all disease sites based on optical-tracking technology and is independent of treatment room, treatment machine, couch top design, immobilization device, or patient internal anatomy. The system

has been well adopted for use in a busy clinical environment because of its seamless workflow and effectiveness in catching setup errors. Due to its advantages of continuous tracking capability, no radiation dose, and fully automated clinical workflow, it serves as an ideal complement to complex IGRT systems in ensuring patient safety in radiotherapy.

Listing 7: User prompt for single LLM-call baselines in long-form patent drafting.

G Supplementary Experimental Results

G.1 Supplementary Ablation Study

In this subsection, we evaluate the robustness of LogicTree-RAG by instantiating the framework with different backbone LLMs, while keeping all other components and hyperparameters unchanged. As shown in Table 4, LogicTree-RAG consistently produces long and well-structured patent drafts across all backbones, indicating that controlling the tree size enables stable and sufficient output length independent of the underlying model. Content-level metrics such as coverage and factuality remain stable, while language-level metrics, including style and repetition rate, gradually improve with stronger backbones. Notably, when both LogicTree-RAG and COPGEN [16] are instantiated with the same Llama-3-8B backbone, LogicTree-RAG achieves substantially better performance and approaches the performance of the SFT-enhanced COPGEN variant, highlighting the effectiveness of the proposed logic-centric generation framework without relying on task-specific fine-tuning. Overall, these results demonstrate that the gains of LogicTree-RAG stem primarily from its structured generation paradigm rather than dependence on a particular backbone model.

Table 4: Performance of LogicTree-RAG with different backbone LLMs.

Method	Output Tokens	Content-level				Language-level			Time (s) / Patent (Token)
		Coverage $\uparrow$	Factuality $\uparrow$		Semantic Similarity $\uparrow$	Style $\uparrow$	Repetition Rate $\mathcal{R}_r \downarrow$	Coherence $\uparrow$	
			$\mathcal{F}_{Pat}$	$\mathcal{F}_{Src}$					
Heuristic Skyline									
Reference Patent (GT)	18.16k	89.05	88.74	88.80	100	100	13.91	100	-
Source Paper	8.03k	44.77	46.06	88.68	39.37	39.26	8.39	98.60	-
Outline (Long)	1.43k	39.11	61.79	61.93	10.16	23.22	20.14	85.69	-
LogicTree-RAG									
Llama3-8B [62]	16.25k	40.25	56.70	61.95	35.74	55.89	7.16	96.98	209 (12.86)
Qwen2.5-72B [64]	17.82k	42.08	57.42	66.09	34.89	62.28	5.87	97.56	260 (14.58)
DeepseekV3 [31]	19.15k	43.20	58.21	64.83	37.95	58.24	5.86	97.10	320 (16.69)
Qwen3-80B [30]	20.22k	43.79	58.92	66.34	38.91	65.68	4.14	97.45	311 (15.38)

G.2 Supplementary Study on Agentic Planning

To directly evaluate the role of structured planning, we compare LogicTree-RAG with two representative agentic paradigms, ReAct [36] and Plan-then-Execute [37]. To ensure a fair comparison, we expose our core modules (i.e., semantic chunking, node-aware retrieval, and evidence-guided generation) as callable tools, and replace the logic-tree-based orchestration with generic agent planning strategies. This design isolates the effect of explicit structural planning (logic tree) versus generic agentic decomposition, while keeping underlying capabilities comparable.

As shown in Table 5, despite using the same underlying tools, the agentic variants exhibit lower coverage, weaker factual grounding, and higher repetition, indicating incomplete and less controlled generation. In contrast, LogicTree-RAG achieves more balanced and consistent outputs, with superior performance across both content-level and language-level metrics. These results suggest that generic agentic planning alone is insufficient for long-form structured generation. Instead, explicit global structure modeling via the logic tree is critical for coordinating content organization, evidence grounding, and generation control.

G.3 Evidence-support Analysis

To assess retrieval-grounded generation at the evidence level, we evaluate whether generated content is supported by retrieved evidence for each retrieval-triggered node, using an LLM-as-judge (GPT-5) and human audit. Each case is categorized as fully supported, partially supported, or unsupported. We further conduct human auditing

Table 5: Performance comparison of LogicTree-RAG against agentic baselines for patent drafting.

Method	Output Tokens	Content-level				Language-level			Time (s) / Patent (Token)
		Coverage $\uparrow$	Factuality $\uparrow$		Semantic Similarity $\uparrow$	Style $\uparrow$	Repetition Rate $\mathcal{R}_r \downarrow$	Coherence $\uparrow$	
			$\mathcal{F}_{Pat}$	$\mathcal{F}_{Src}$					
LogicTree-RAG	20.22k	43.79	58.92	66.34	38.91	65.68	4.14	97.45	311 (15.38)
$\hookrightarrow$ w/ ReAct	16.86k	40.28	55.98	64.85	36.86	65.52	10.24	96.18	336 (19.93)
$\hookrightarrow$ w/ Plan-Execute	18.58k	42.02	56.35	64.96	35.91	64.97	8.95	96.59	325 (17.49)

on 20 randomly sampled test cases under the same criteria. Results are summarized in Table 6. LogicTree-RAG achieves a substantially higher proportion of fully supported content and fewer unsupported cases than COPGEN, indicating stronger attribution faithfulness and more effective use of retrieved evidence. Human auditing yields consistent trends, confirming that retrieved evidence is actively used to support generation rather than being passively retrieved. These findings align with the ablation results in Table 2, where removing evidence-grounded refinement σ_{R2} consistently degrades performance, while removing internal refinement σ_{R3} has a negligible impact. These results suggest that the gains of LogicTree-RAG primarily arise from improved evidence-grounded generation, rather than increased pipeline complexity or auxiliary mechanisms.

Table 6: Evidence-support analysis of retrieved evidence and generated content.

Method	Fully (%)	Partial (%)	Unsupported (%)
LLM-as-judge			
COPGEN	70.4	20.2	9.4
LogicTree-RAG	85.6	12.0	2.4
Human Audit			
COPGEN	60	30	10
LogicTree-RAG	75	25	0

G.4 Computational Cost Analysis

To improve transparency, we report a detailed cost breakdown of LogicTree-RAG along the workflow pipeline. As shown in Table 7, the overall cost is well controlled, with a balanced distribution across stages. In particular, recursive logic tree construction introduces only moderate overhead, while the majority of cost arises from traversal and generation. Importantly, the recursion depth is explicitly bounded by tree budget hyperparameters (see Appendix F.3), preventing uncontrolled expansion and ensuring a favorable quality-efficiency trade-off rather than gains driven by excessive API usage.

Table 7: Supplementary computational cost analysis.

Computational Cost (/patent)	Cost Breakdown
LLM Calls: 20	chunking&tree init 5%, tree construction 45%, traversal 50%
Runtime: 311s	chunking&tree init 16%, tree construction 39%, traversal 45%
Total Input Tokens: 7.60k	chunking&tree init 29%, tree construction 37%, traversal 34%

G.5 Human Expert Study

We conduct a human expert evaluation with three patent practitioners to assess the practical quality of generated patents. We randomly select 10 samples from the test set. For each sample, practitioners are provided with the source paper, the generated patent, and the reference patent, along with the corresponding outlines (long version) to ensure fair comparison. Experts perform pairwise comparisons (win/loss/tie) between LogicTree-RAG and each baseline. We adopt five evaluation criteria covering both legal correctness and technical quality, as defined in Listing 8. All evaluations are conducted in a blinded manner, where the identities of methods are anonymized.

The win-rate results are summarized in Table 8. LogicTree-RAG consistently outperforms all baselines across evaluation criteria, achieving strong human preference in overall quality and technical correctness. In particular, LogicTree-RAG shows substantial advantages in Claim Structure and Antecedent Basis, indicating improved structural coherence and consistency in long-form patent drafting. Gains in Fidelity further demonstrate that the generated patents remain closely grounded in the source content.

```

1 ### Expert Evaluation Criteria
2 * Overall Quality: overall usefulness as a patent draft for practitioner revision
3 * Legality: conformity to basic patent drafting and legal-form requirements
4 * Claim Structure: clarity and correctness of claim organization and dependencies
5 * Antecedent Basis: consistency of terminology and antecedent basis
6 * Fidelity: faithfulness to the technical content of the source paper

```

Listing 8: Definitions of evaluation criteria used in human expert assessment.

Table 8: Win-rate results of LogicTree-RAG in human expert evaluation.

Method	Overall	Legality	Claim Structure	Antecedent Basis	Fidelity
Ours vs Qwen3-80B	75.7	69.8	84.6	84.8	88.5
Ours vs DeepseekV3	84.4	65.1	88.5	90.1	90.2
Ours vs LongWriter	78.8	76.5	90.6	92.0	85.9
Ours vs COPGEN	70.5	65.7	85.2	82.1	78.5

G.6 Sensitivity to Tree Growth Budget

We further analyze the sensitivity of LogicTree-RAG to the tree growth budget by varying the maximum tree depth and the maximum number of tree nodes, while keeping all other components and hyperparameters fixed. As shown in Figure 8, increasing the tree budget consistently improves coverage, indicating that deeper or broader trees enable more comprehensive exploration of the document structure. However, the gains diminish beyond moderate budgets, suggesting that additional expansions primarily capture long-tail details. In particular, the default configuration, i.e., maximum depth = 3 and maximum number of nodes = 64, lies in this favorable regime, achieving high coverage with reasonable efficiency. In contrast, token efficiency decreases as the tree budget grows, due to the increased retrieval and generation overhead incurred by additional nodes. These results reveal a clear quality–cost trade-off and demonstrate that LogicTree-RAG allows controllable scaling, where moderate tree budgets achieve a favorable balance between coverage and efficiency.

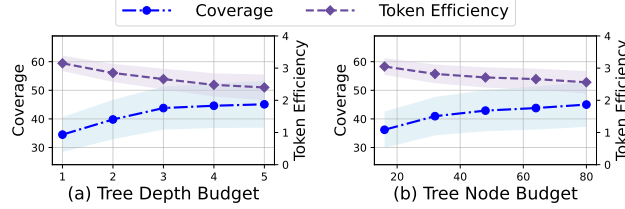


Figure 8: Sensitivity to tree growth budget in LogicTree-RAG.

G.7 Case Study

To qualitatively assess the structural fidelity of long-form patent generation, we present a case study comparing a patent generated by our method with its ground-truth counterpart, as shown in Figure 9 and 10. Both documents are visualized using section-level color coding to facilitate direct comparison of document organization and length distribution. As observed, the generated patent closely matches the reference patent in overall length and preserves a well-balanced allocation across major sections.

BACKGROUND OF THE INVENTION Wound healing and solid tumor progression share fundamental biological features, particularly increased vascular permeability associated with inflammation. In response to injury or infection, vascular barriers in wounds are transiently disrupted, enabling extravasation of immune cells, plasma proteins, and inflammatory mediators. These features underlie the enhanced and well-characterized physiological response that facilitates tissue repair. Solid tumors similarly exhibit abnormal and persistently inflated vasculature and have therefore been described as non-healing wounds. Tumor-associated blood vessels are structurally irregular and functionally compromised, resulting in increased permeability and impaired clearance. These features underlie the enhanced permeability and retention (EPR) effect, which allows macromolecular agents, including nanoparticles and antibody-based therapeutics, to preferentially accumulate within tumor tissue. Although this effect has been utilized to improve tumor-selective drug delivery, its magnitude and reliability vary significantly across tumors and patients. Emerging evidence indicates that inflammatory stimuli within the tumor microenvironment can substantially amplify vascular permeability beyond baseline EPR levels. In particular, tumor-associated inflammation induced by microbial agents or pro-inflammatory cytokines has been shown to promote non-specific accumulation of macromolecules within tumors. The present invention is based on the recognition that controlled modulation of tumor vascular permeability using inflammatory or vasculature-active agents provides a broadly applicable strategy for enhancing the delivery and efficacy of macromolecular and nanoparticle-based agents in solid tumors.	Detailed Description of Embodiments The present invention will be further described in detail below with reference to the accompanying drawings and embodiments, but the present invention is not limited to the following specific embodiments. ① A novel algorithmic protocol is implemented to dynamically modulate tumor vascular permeability by inducing a localized inflammatory cascade through the controlled system of intravenous administration of a pro-inflammatory cytokine, specifically tumor necrosis factor-α (TNF-α), at a precise concentration of 10-50 ng/kg body weight, delivered intravenously via a programmable infusion pump calibrated to maintain a plasma half-life of 8-12 minutes, thereby triggering endothelial junctional reorganization via RhoA/ROCK-mediated actomyosin contraction and VE-cadherin internalization within tumor vasculature while sparing non-tumorous tissues through differential expression of TNF receptor subtype 1 (TNFR1) on tumor endothelial cells, which exhibit 3-7-fold higher surface density compared to normal endothelium as quantified by flow cytometry and confirmed by immunohistochemical staining of CD31+ vessels. ② The system integrates a time-synchronized nanoparticle delivery module that initiates intravenous injection of PEGylated liposomal doxorubicin (Doxil) or polymeric nanoparticles with hydrodynamic diameters of 80-120 nm precisely 72 hours after TNF-α administration, a temporal window validated by longitudinal interval microscopy to coincide with peak vascular permeation and reduced interstitial fluid pressure, wherein nanoparticle extravasation is enhanced by 4.2-fold relative to uninduced controls due to the formation of 150-400 nm transient endothelial gaps stabilized by fibronectin deposition and reduced pericyte coverage, as confirmed by multiphoton imaging of FITC-dextran (70 kDa) leakage and confocal analysis of α-SMA+ pericyte detachment. ③ An adaptive targeting algorithm employs real-time biotransformation feedback derived from near-infrared fluorescence-labeled nanoparticles and gamma-counting of radiolabeled (99mTc) albumin to dynamically adjust the nanoparticle infusion rate and volume based on tumor-to-blood fluorescence intensity ratios exceeding 4.5:1, ensuring that nanoparticle accumulation is gated not by passive diffusion alone but by the presence of TNF-α-induced vascular remodeling, with algorithmic thresholds triggering cessation of infusion if tumor fluorescence intensity plateaus or declines, thereby preventing systemic overload and minimizing off-target accumulation in liver and spleen as validated by ex vivo organ bioluminescence. ④ A spatially resolved delivery optimization engine is embedded to correlate nanoparticle accumulation with hypoxic regions identified by paramagnetic contrast agents and hypoxia-inducible factor-1α (HIF-1α) immunofluorescence, ensuring that nanoparticle penetration is prioritized along vascular corridors adjacent to necrotic cores, where TNF-α-induced permeability is sustained for up to 96 hours due to suppressed VEGF-A clearance and persistent macrophage infiltration, thereby enabling uniform drug distribution throughout the tumor periphery as quantified by digital histomorphometry of doxorubicin nucleofluorescence across 500 $\times$ histological sections per tumor. ⑤ The system incorporates a feedback-controlled cytokine priming module that administers TNF-α dosing frequency and magnitude based on pre-treatment tumor vascular density measured by dynamic contrast-enhanced MRI (DCE-MRI) with Gd-DTPA, wherein tumors with baseline K_{trans} values below 0.05 min⁻¹ receive a priming dose of 50 ng/kg TNF-α followed by a second dose at 72 hours, wherein tumors with K_{trans} above 0.08 min⁻¹ receive a single 10 ng/kg dose, preventing vascular collapse and ensuring sustained permeability without inducing systemic toxicity, as monitored by serum IL-6 and MCP-1 levels maintained below 50 pg/mL and 100 pg/mL, respectively, throughout the treatment window. The specific process of the said step2a is: ②-1 The nanoparticle delivery module employs a deterministic temporal trigger anchored to a synchronized real-time clock calibrated to the precise moment of TNF-α bolus administration, ensuring that intravenous infusion of PEGylated liposomal doxorubicin or polymeric nanoparticles commences exclusively at the 72-hour mark. This temporal precision is maintained within a ± 30-minute window (71.5-72.5 hours) to exclude suboptimal permeability states observed earlier or later, as established through longitudinal interval microscopy of tumor-bearing mice imaged at 6-hour intervals over 96 hours. The 72-hour window corresponds to the peak of vascular permeation, marked by maximal FITC-dextran (70 kDa) flux, where the effective endothelial barrier thickness declines by 68% compared to baseline, directly correlating with the onset of maximal interstitial accessibility for nanoparticles of 80-120 nm hydrodynamic diameter. ②-2 Each nanoparticle batch undergoes rigorous hydrodynamic characterization prior to administration using quasi-static light scattering via a Malvern Zetasizer 3000, with acceptance criteria requiring a mean diameter of 97 $\pm$ 5 nm and a polydispersity index below 0.15 to ensure compatibility with the 150-400 nm endothelial gaps formed by TNF-α. PEGylated liposomal doxorubicin is synthesized through thin-film hydration of HEPES-Chol-DSPE-PEG2000 at a 20:45:1 molar ratio, followed by extrusion through dual 0.1 μm Nuclepore filters using a Lipex Thermomixer. Extruder generates single unilamellar vesicles with uniform size distribution. Polymeric nanoparticles are produced via optimized solvent evaporation kinetics, ensuring identical size constraints. Any batch failing to meet dimensional specifications is automatically quarantined and flagged for re-synthesis, eliminating the risk of non-compliant particles entering the delivery pathway. ②-3 Delivery is contingent upon the concurrent presence of fibronectin deposition and pericyte detachment within the tumor vasculature adjacent to the injection site, as quantified by confocal microscopy of tumor sections immunostained for fibronectin and α-SMA. A threshold-based detection logic mandates a minimum fibronectin signal intensity of 2.1-fold above pre-TNF-α baseline and a reduction in α-SMA+ pericyte coverage to less than 30% of initial levels, determined through pixel intensity segmentation across fluorescence channels. If these biomarkers are not satisfied at the 72-hour trigger point, the system delays injection in 30-minute increments, up to two additional assessments, with real-time re-evaluation of endothelial structural markers. Should the thresholds remain unmet after 72.5 hours, the system terminates the delivery sequence and logs a protocol deviation, ensuring that nanoparticle extravasation occurs only when the endothelial microenvironment is structurally primed. ②-4 The system incorporates a real-time feedback validation loop that empirically confirms the 4.2-fold enhancement in nanoparticle extravasation by analyzing multiphoton imaging data of FITC-dextran (70 kDa) leakage within the tumor interstitium during the first 10 minutes post-injection. The extravasation rate is computed as the temporal slope of fluorescence intensity increase, and if this rate falls below 1.5$\times$ the baseline observed in TNF-α-naïve tumors, the system immediately halts further infusion and triggers a manual review alert. This validation is independent of pharmacokinetic or biotransformation metrics and is exclusively tied to the dynamic endothelial state defined by transient gap formation and pericyte withdrawal, ensuring that therapeutic delivery efficacy is not extrapolated but directly measured in vivo, thereby ensuring strict mechanistic alignment with the validated biological window of enhanced permeability. The specific process of the said step2a is: ②-3 The algorithm continuously acquires synchronized dual-modality biotransformation data by integrating near-infrared fluorescence imaging of IRDye800CW-labeled PEGylated liposomal doxorubicin with real-time gamma-counting of 99mTc-labeled albumin serum albumin. Fluorescence signals are quantified as mean pixel intensity within a tumor region of interest defined by co-registered CT anatomical segmentation, while gamma counts are normalized to blood activity measured via jugular vein sampling at 5-second intervals. The tumor-to-blood ratio (TBR) is calculated every 30 seconds using the formula $TBR = \frac{(Fluorescence_{tumor} / Fluorescence_{blood}) \times (Gamma_{tumor} / Gamma_{blood})}{TBR_{baseline}}$, wherein Fluorescence_{tumor} and Fluorescence_{blood} represent the mean NIR intensity in the tumor and contralateral deltoid muscle, respectively, and Gamma_{tumor} and Gamma_{blood} denote decay-corrected background-subtracted counts per second from the tumor and blood regions. Infusion is initiated only after three consecutive TBR measurements exceed 4.5:1, confirming sustained TNF-α-mediated vascular permeability beyond baseline EPR effects. ②-2 Once infusion commences, a proportional-integral-derivative (PID) control loop modulates the infusion rate in real time, with the error term defined as the deviation of the current TBR from the 4.5:1 target threshold. The integral component accumulates cumulative deviation over a 5-minute sliding window to mitigate overshoot and ensure stable accumulation, while the derivative term dampens transient signal fluctuations induced by respiratory motion or pulsatile perfusion. The infusion pump adjusts flow rate in 0.05 mL/min increments between 0.1 mL/min and 2.0 mL/min, strictly bounded by a maximum safe volume limit of 1.5 mL/kg body weight. A 16-point moving average filter is applied to the TBR time series to suppress noise from photon shot noise and tissue autofluorescence drift, ensuring that rate adjustments respond only to persistent, biologically meaningful changes in tumor accumulation rather than stochastic artifacts. ③-1 Infusion is terminated under three distinct algorithmic conditions: if the TBR remains below 4.5:1 for five consecutive scans via imaging intervals, indicating failure to achieve a TNF-α-dependent vascular remodeling; if tumor fluorescence intensity declines by more than 15% over a 10-minute window, signaling saturation or clearance of extravasated nanoparticles; or if the TBR exhibits a slope of less than 0.02 per minute for 10 consecutive minutes, reflecting exhaustion of vascular permeability capacity. Upon termination, the system logs the total infused volume, infusion duration, and final TBR, and automatically initiates a post-treatment biotransformation protocol. Ex vivo gamma counting and fluorescence spectroscopy are performed on harvested liver, spleen, kidneys, and muscle tissues to quantify off-target accumulation, confirming that systemic exposure is minimized and that nanoparticle retention is confined to the target site. ③-4 The algorithm enforces a hard cap of 30 minutes on total infusion duration, irrespective of TBR trajectory, to prevent prolonged systemic circulation that could lead to reticuloendothelial system sequestration. This limit is calibrated to the pharmacokinetic half-life of 99mTc-albumin (~ 15 minutes) and the photophysical decay kinetics of IRDye800CW, including its quantum yield of 0.18 and molar extinction coefficient of 250,000 M⁻¹cm⁻¹, ensuring that termination occurs prior to significant hepatic or splenic uptake driven by Kupffer cell internalization. All thresholds are empirically validated against histological and quantitative biotransformation data from murine tumor models, demonstrating that cessation criteria reliably prevent off-target accumulation while maximizing tumor-specific delivery under dynamic vascular conditions.
SUMMARY OF THE INVENTION In one embodiment, the present disclosure relates to a method for enhancing the accumulation of functional agents within solid tumor tissue. The method comprises administering to a subject diagnosed with a solid tumor a carrier-based delivery composition selected from nanoparticles or antibody-based targeting moieties, wherein the carrier composition is configured to transport a therapeutic agent for cancer treatment or an imaging agent for tumor detection. The method further includes administering a permeability-modulating factor capable of transiently altering tumor-associated vasculature. The permeability-modulating factor may be selected from bacterial agents or derivatives thereof, inflammatory mediators, or cytokine-like stimulating compounds. As a result, delivery of the functional agent to the tumor microenvironment is increased relative to administration in the absence of the permeability-modulating factor, thereby improving tumor localization efficiency. In certain implementations, the disclosed method increases the preferential distribution of the functional agent to tumor tissue compared to systemic circulation. Specifically, the ratio of agent accumulation within the solid tumor relative to that present in the bloodstream is elevated when the permeability-modulating factor is co-administered. Such selective enhancement enables improved therapeutic efficacy or diagnostic sensitivity while reducing off-target exposure. In another aspect, the invention provides a kit comprising one or more containers that include a permeability-modulating factor and a carrier-based delivery composition. The delivery composition may be a nanoparticle or antibody configured to transport a therapeutic anti-neoplastic compound or a detectable imaging payload. The permeability-modulating factor may be provided in the same or a separate container and is selected from bacterial components, pro-inflammatory signaling molecules, or agents that destabilize microvessel structures. In a further aspect, a pharmaceutical composition is disclosed that combines a permeability-modulating factor with a nanoparticle or antibody carrying a therapeutic or imaging agent. The compositions and methods described herein collectively enable improved targeting, detection, and treatment of solid tumors. Additional features, variations, and advantages of the invention will be apparent to those skilled in the art upon consideration of the present disclosure.	

Figure 9: Qualitative case study illustrating a patent example generated by LogicTree-RAG.

[illegible]

Figure 10: Qualitative case study illustrating the ground-truth reference patent.

H Discussion on Human Evaluation and Legal Adequacy

While this work focuses on algorithmic mechanisms for controllable, evidence-grounded long-form patent drafting in an early-stage drafting setting, we emphasize that the proposed framework is intended as a structured drafting assistant rather than a replacement for professional patent attorneys. Accordingly, aspects critical to real-world patent practice, such as legal adequacy, claim dependency correctness, and antecedent basis consistency, ultimately require expert human judgment. These criteria extend beyond surface-level textual quality and factual alignment, involving nuanced legal interpretation and jurisdiction-specific drafting conventions that remain difficult to reliably capture with automatic metrics.

We view human evaluation by patent practitioners as an essential direction for future validation. A principled evaluation protocol would involve expert assessment of generated drafts along dimensions including (i) legal adequacy of technical disclosure, (ii) correctness and consistency of claim dependencies, and (iii) validity of antecedent basis usage, potentially complemented by qualitative feedback on drafting clarity and completeness. We believe that the logic-tree-based intermediate representation introduced in this work provides a favorable foundation for such evaluations by explicitly exposing conceptual dependencies and evidence grounding, thereby facilitating systematic inspection and correction by human experts.